

Bioinspiration & Biomimetics



PAPER

A two-dimensional hydrodynamics prediction framework for mantle-undulated propulsion robot using multiple proper orthogonal decomposition and long short term memory neural network

RECEIVED
4 September 2023

REVISED
26 October 2023

ACCEPTED FOR PUBLICATION
17 November 2023

PUBLISHED
29 November 2023

Zixiang Ying¹ , Haozhi Zhang¹, Linxiang Wang^{1,*} and Roderick Melnik²

¹ State Key Laboratory of Fluid Power and Mechatronic Systems, Zhejiang University, 310027 Hangzhou, People's Republic of China

² MS2Discovery Interdisciplinary Research Institute, Wilfrid Laurier University, Waterloo N2L 3L5, ON, Canada

* Author to whom any correspondence should be addressed.

E-mail: wanglx236@zju.edu.cn

Keywords: multiple feature identification, proper orthogonal decomposition, long short-term memory neural networks, mantle-undulated propulsion robot, bionic propulsion

Abstract

In this paper, a deep learning based framework has been developed to predict hydrodynamic forces on a mantle-undulated propulsion robot (MUPRo). A multiple proper orthogonal decomposition (MPOD) algorithm has been proposed to efficiently identify fluid features near the undulating mantle of the MUPRo globally and locally. The results indicate that the L^2 error of the solution states near the undulating boundary of the proposed MPOD algorithm converges almost linearly to 0.2%. Furthermore, a hydrodynamics prediction framework has been developed based on the proposed MPOD algorithm, where a long short-term memory neural network predicts the temporal coefficients of the MPOD spatial modes. The developed framework achieves economical and reliable predictions of hydrodynamic forces acting on the undulating boundary compared to simulations and experiments. Moreover, the L^2 error of the developed framework is one to two orders of magnitude lower than that of the frameworks based on the classical POD algorithm when the degrees of freedom are consistent. Finally, the reliability of the proposed MPOD-NIROM is discussed through an offline parameter planning case of an aquatic-inspired robot. The model presented in this paper can provide support for the offline parameter planning of aquatic-inspired robots.

1. Introduction

Aquatic organisms utilise their bodies for stability and manoeuvrability under specific fluid flow conditions characterised by the Reynolds number ($Re \leq 10^4$) [1–3]. For instance, swimmers employ body/caudal fin undulations to generate propulsion, utilising the reverse travelling wave motion with maximum amplitude at the tail. This method is widely adopted in carangiform swimming, which is prevalent in most swimmers [4, 5]. However, lampreys and eels employ anguilliform swimming [6–8], while tuna use thunniform swimming [9]. Jellyfish use propulsion modes based on periodic momentum injection. For example, oblate jellyfish use the rowing propulsion

[10, 11], whereas prolate jellyfish utilise jet propulsion technique [12–15]. Some propulsion techniques have been shown to be efficient and attractive for designing underwater vehicles [16]. In particular, geometrically symmetric jellyfish-inspired jet-propelled underwater robots exhibit excellent orientation [17–19].

The exploration and discussion of jellyfish propulsion can provide guidance in the control of propulsion and optimisation of parameters in a jellyfish-inspired robot [20]. The hydrodynamic mechanisms of jellyfish, as well as the jellyfish-inspired robots, have garnered significant attention from the scientific community through experimental measurements [12, 15, 21–23], numerical simulations [24–28], and reduced-order models [29].

From a hydrodynamic perspective, the propulsion of a jellyfish-inspired robot is a canonical fluid-structure interaction problem, and the feature of the reversed Kármán vortex street is explicitly included in the solution states of the two-dimensional computational fluid dynamics (CFD) model [30]. The two-dimensional CFD model is also referred to as the high-fidelity model (HF model) in the subsequent sections. The high computational cost makes the design or control of the jellyfish-inspired robot challenging based on the HF model.

One approach for decreasing the computational burden involved in forecasting when predicting jellyfish-mimicking robotic propulsion is to decrease the order of the full-order HF model using the proper orthogonal decomposition (POD) algorithm. This is done while taking into consideration the energetic convergence properties of the POD algorithm with regards to feature identification [31–33]. The POD algorithm is extensively employed to capture the fluid's dynamic characteristics [3, 34–36], and research indicate that it is a suitable method for extracting vortex street structures [37–39].

When discussing the instantaneous hydrodynamic force acting on a jellyfish-inspired robot, it is expected that fluid features near the robot can be reliably predicted. However, the identification of fluid features in the vicinity of the robot poses a challenge if the classical POD algorithm is used because the solution states in close proximity to the boundary have a low contribution to the total system energy and tend to be ignored by POD-based reduced-order models (ROMs) [29]. To address this issue, a multiple POD (MPOD) algorithm is proposed in this paper for the identification of fluid features near the robot. Drawing inspiration from the multiple numerical simulation algorithms, the MPOD algorithm identifies fluid features at both global scale (i.e. the energy favoured characteristics of the solution states within the computational domain) and local scale (i.e. the crucial characteristics of the region near the moving boundary but lack of energy dominance). First, the fluid features in the solved states are globally identified by the classical POD algorithm, while the unrecognized part of the solution states constitutes the residuals. Subsequently, local feature identification task for the residual is performed to predict the temporal evolution of fluid features around the robot more accurately. To ensure efficient identification of fluid features, the residual is weighted by assigning higher weights to the solution states in close proximity to the robot before performing local feature recognition. In recent years, the weighted POD algorithm (WPOD) has gained popularity [40–44], and studies have shown that this methodology requires fewer POD modes to approximate the solution states than the classical POD methodology [45, 46]. Specifically, when performing the WPOD, solution states extracted from the HF model are weighted

so that snapshots are concentrated in designated regions of the state space [41–45].

However, it is essential to emphasise that the proposed MPOD algorithm generates two sets of POD spatial modes for the ROM, namely global POD spatial modes and local POD spatial modes, and they are not orthogonal to each other. Therefore, it poses a challenge to construct a ROM by performing the Galerkin projection of the governing equations of the full-order HF model onto the solution states described by the POD spatial modes. The classic POD-based ROM details are discussed in the existing literature [31, 32, 38, 47–50]. In recent years, non-intrusive models based on machine learning (ML) have been proposed, which have significantly enhanced the stability and computational efficiency of the POD-based ROM [29, 51, 52]. Furthermore, studies have demonstrated that non-intrusive ROMs (NIROMs) based on ML produce outcomes similar to those of the POD-based reconstructions [51, 53, 54]. Notably, the long short-term memory (LSTM) neural network (NN)-based NIROM can learn and predict sequential problems [55, 56]. Additionally, based on the LSTM NN, NIROM learns the underlying physical dynamics in the full-order HF model solution, making it possible to predict the POD temporal coefficients.

In this paper, the MPOD-NIROM algorithm is proposed as an extension to the posteriori ROM framework. This novel algorithm consists of three main components: (1) multiple feature identification of the solution states, (2) online prediction (which means that the model makes real-time predictions based on new inputs) of the temporal evolution of fluid features and (3) estimation of the hydrodynamic force acting on the robot. The feature identification process is conducted twice, at different scales, namely the global feature identification for the solution states and local feature identification for the residual generated by the global feature identification. The scale reduction of fluid features at the local scale is achieved by weighting the residual with a Gaussian decay function, which smooths the features at the edge. The LSTM-based NIROM predicts the temporal evolution of fluid features online, taking the specified boundary motion describing the robot's undulation as input. Additionally, the hydrodynamic force acting on the robot is estimated based on the predicted solution states. Furthermore, the efficacy of feature identification is evaluated with greater precision, considering the convergence speed of the error. The efficiency and truncation error of the proposed MPOD algorithm are compared to those of the classical POD algorithm, for the cases and experiments discussed in this paper. Additionally, the potential of the developed MPOD-NIROM framework in predicting the hydrodynamic force acting on the robot is demonstrated through an offline parameter planning (i.e. planning the motion-related parameters of the

robot before it performs propulsion) of the mantle-undulated propulsion robot (MUPRo).

The remainder of this paper is presented as follows. The investigated case is depicted in section 2.1, specifically, the prediction of hydrodynamic forces acting upon the undulating mantle of the robot. In section 2.2, the MPOD algorithm proposed is described, which serves as a foundation for the developed MPOD-NIROM framework in section 2.3. Simulations and experimental results as well as detailed discussion are presented in section 3. The effectiveness of the proposed MPOD algorithm in identifying fluid characteristics around the undulating mantle is more accurately evaluated and compared with the classical POD algorithm in sections 3.2 and 3.3. Moreover, in sections 3.4–3.6, the potential of the proposed MPOD-NIROM framework for solution state prediction and its reliability for hydrodynamic estimation are assessed in terms of prediction error and computational cost. An offline parameter planning of the MUPRo based on the proposed MPOD-NIROM is discussed in section 3.6. Finally, in section 4, conclusions are drawn, limitations are discussed and potential extensions of the developed MPOD-NIROM framework are outlined. Appendices A and B provide detailed introductions to the HF model and LSTM NN, respectively.

2. Methodology

In this paper, we propose the MPOD algorithm to enable the economic and reliable prediction of hydrodynamic forces acting on a mantle. Based on this algorithm, we developed the MPOD-NIROM framework. Our proposed MPOD algorithm builds upon the classic POD algorithm and aims to identify efficiently the features of the unsteady fluid surrounding the undulating mantle of the robot. This identification enables a reliable prediction of the hydrodynamic force acting on the undulating mantle based on the ROM with lower degrees of freedom (DoF, which refers to the expansion degree of POD spatial modes when construct the ROM). In this section, we introduce the MPOD algorithm and the MPOD-NIROM framework that we have developed.

2.1. Geometric assumption and high fidelity modelling

The present study aims to predict the start-up propulsion of the jellyfish-inspired MUPRo. Inspired by jellyfish, MUPRo is designed with symmetrical geometry and symmetrical propulsion for better orientation. To simplify the experiments, we assume a HF model based on 2D computational domain. The 2D prototype is depicted in figure 1(A), while the computational domain is presented in figure 1(B). Moreover, we applied axisymmetric assumptions in the HF model and the subsequent ROM to reduce the

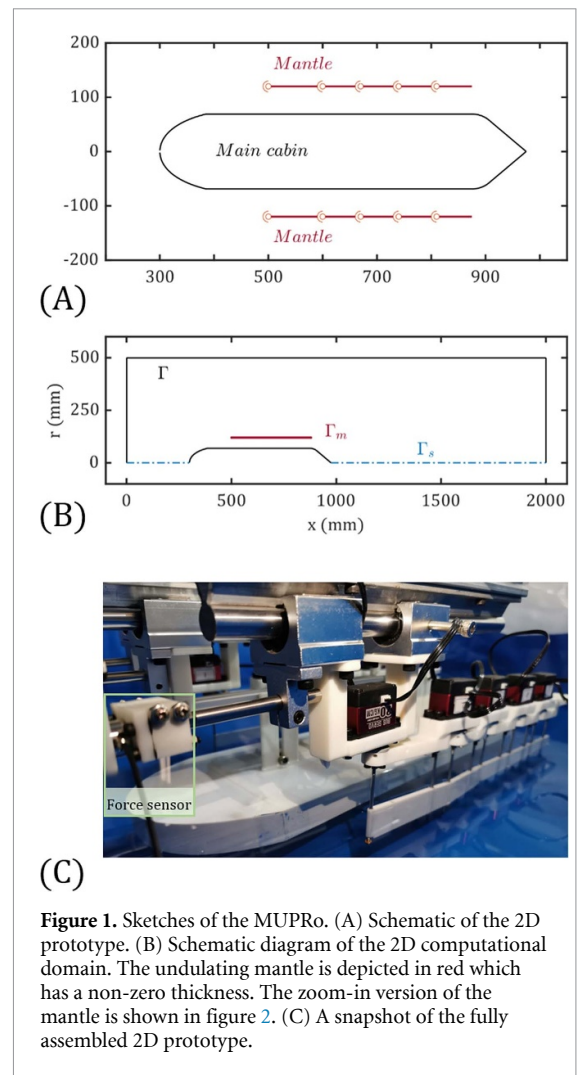


Figure 1. Sketches of the MUPRo. (A) Schematic of the 2D prototype. (B) Schematic diagram of the 2D computational domain. The undulating mantle is depicted in red which has a non-zero thickness. The zoom-in version of the mantle is shown in figure 2. (C) A snapshot of the fully assembled 2D prototype.

number of elements required for the simulation to converge, since both the computational domain and the undulating motion of the mantle are symmetric about the x -axis. However, it is worth mentioning that the lofting result for the 2D computational domain, where the rotational symmetry assumption is applied, does not entirely agree with the 3D prototype due to the neglect of the connection between the mantle and the main cabin [29].

Studies have revealed that the size of a computational domain affects the solution accuracy of unsteady problems [57–59]. Based on this, the size of the computational domain is strictly consistent with the size used in the 2D prototype experiments (figure 1(C)), the details can be find in the recent work [29]. As shown in figure 1(B), the size of the 2D numerical pool is 2000 mm $\times$ 500 mm, the mantle starts at (500 mm, 120 mm), and the front of the main cabin is located at (300 mm, 0). The length and width of the main cabin are 672.48 mm and 140 mm respectively, and the length and thickness of the mantle are 380 mm and 5 mm respectively.

In the HF model, the undulation of the MUPRo is explicated as the displacement of the grid on the Γ_m .

The kinematics for the body axis of the mantle are formulated in the form of the retrograde-travelling wave postulated by Lighthill [60],

$$Y(\xi, t) = A(\xi) \sin(2\pi k\xi - \sigma t), \quad (1)$$

$$A(\xi) = A_0 + A_1\xi + A_2\xi^2, \quad (2)$$

where $\xi = x - 500\text{mm}$ is the measured length along the body axis of the mantle, $k \in [0.1, 0.8]$ is the dimensionless wavenumber, $\sigma = 2\pi/T$ is the angular frequency, $T \in [1.5, 3.0]$ is the undulating period. $A(\xi)$ is the coefficient of the first term of the Fourier series defining the maximum radial displacement of the mantle. In the present study, the amplitude at the start point of the mantle is 0, and the amplitude at the end of the mantle is 0.1 times the mantle length (380 mm), and the amplitude is linearly excessive along the mantle axis, i.e. $A_0 = 0$, $A_1 = 0.1$, $A_2 = 0$.

The HF model presented in this work is constructed using the FreeFem++ software [61]. The model utilises the incompressible Navier–Stokes equations under arbitrary Lagrangian–Eulerian (ALE) descriptions as the governing equations. To perform temporal integration of the momentum equation, the model employs the fractional step formulation. Spatial discretisation of the governing equations is accomplished by employing streamline upwind Petrov–Galerkin-based FEM. Grid motion within the computational domain is addressed using the modified Laplace algorithm. Further details (e.g. governing equations and boundary conditions) are presented in appendix A. The validation study of the HF model is described in detail in section 3.1.

2.2. MPOD

The current algorithm is achieved a posteriori, where the solution states s_i ($[s_1, s_2, s_3]^T = [u, v, p]^T$) required for the proposed MPOD algorithm are extracted from the HF model with specified parameter combinations. The matrix of the solution state vector s is defined as follows,

$$\mathbf{s} = \begin{bmatrix} s_1 \\ s_2 \\ s_3 \end{bmatrix} = \begin{bmatrix} u \\ v \\ p \end{bmatrix}, \quad (3)$$

where u is the velocity of the fluid in the x -direction, v is the velocity of the fluid in the r -direction and p is the pressure. When predicting the hydrodynamic forces acting on the underwater vehicle, the solution states on the boundary $s_i|_{\Gamma_m}$ is critical. To economically predict the hydrodynamic forces exerted on the mantle of the MUPRO, a modified POD algorithm, referred to as Multiple Proper Orthogonal Decomposition (MPOD), has been proposed. Drawing inspiration from multi-scale numerical simulation algorithms, the MPOD algorithm presented in this paper aims to identify fluid features at both global and local scales.

2.2.1. Global feature identification

The snapshot POD technique [31, 62] is utilised to efficiently and numerically approximate the HF model by identifying the primary characteristics of the solution states within the computational domain. The Reynolds decomposition is required to obtain the expansion of the solution states s_i into the temporal mean and the fluctuating component,

$$s_i(\mathbf{x}, t, \boldsymbol{\theta}) = \langle s_i(\mathbf{x}, t, \boldsymbol{\theta}) \rangle + s_i'(\mathbf{x}, t, \boldsymbol{\theta}), \quad (4)$$

where $\langle s_i(\mathbf{x}, t, \boldsymbol{\theta}) \rangle$ denotes the time-averaged term and $s_i'(\mathbf{x}, t, \boldsymbol{\theta})$ denotes the fluctuating component, $t = [t_1, t_2, \dots, t_{n_s}]$ is the time step for extracting snapshots from $3T$ to $5T$ with a time interval $\Delta t_{n_s} = 0.05T$, where n_s denotes the number of collected snapshots. $\boldsymbol{\theta}(k, T) = [k, T]$ refers to the control parameter of MUPRO, which consists of the wave number k together with the undulating period T . Performing the snapshot POD method, the fluctuating component can be approximated by the global energy-favoured spatial mode $\Phi_j^{s_i}(\mathbf{x})$ and corresponding temporal coefficient $a_j^{s_i}(t, \boldsymbol{\theta})$ ($j = 1, 2, \dots, r_1^{s_i}$), where $r_1^{s_i}$ denotes the DoF needed for the global feature identification). The truncation error is expressed as a residual $R(r_1^{s_i}, \boldsymbol{\theta})$,

$$s_i'(\mathbf{x}, t, \boldsymbol{\theta}) = \sum_{j=1}^{r_1^{s_i}} \Phi_j^{s_i}(\mathbf{x}) a_j^{s_i}(t, \boldsymbol{\theta}) + R(r_1^{s_i}, \boldsymbol{\theta}). \quad (5)$$

To intuitively discuss the performance of the proposed MPOD algorithm on feature identification, the convergence rate is defined from the perspective of error reduction,

$$c^{s_i}(j) = (1 - \text{RIC}(j+1))(1 - \text{RIC}^{s_i}(j))^{-1} \quad \text{for } j = 1, 2, \dots, r_1^{s_i}, \quad (6)$$

where $c^{s_i}(j)$ denotes the convergence rate of the relative information content corresponding to j . $\text{RIC}^{s_i}(j)$ denotes the relative information content (RIC) for the global feature identification, which is defined in the previous research [63],

$$\text{RIC}^{s_i}(j) = \sum_{l=1}^j \lambda^{s_i}(l) \left(\sum_{l=1}^{n^{s_i}} \lambda^{s_i}(l) \right)^{-1}, \quad (7)$$

where $\lambda^{s_i}(j)$ denotes the eigenvalue of the eigenvector $\Phi_j^{s_i}$.

2.2.2. Local feature identification

Due to the lack of energy dominance (which implies that the features are almost uniformly distributed over the full range of POD spatial modes and it is difficult to reconstruct them with a few POD spatial modes), the solution states near the moving boundary are not fully characterised. This implies that some crucial fluid features are left unaccounted for in the

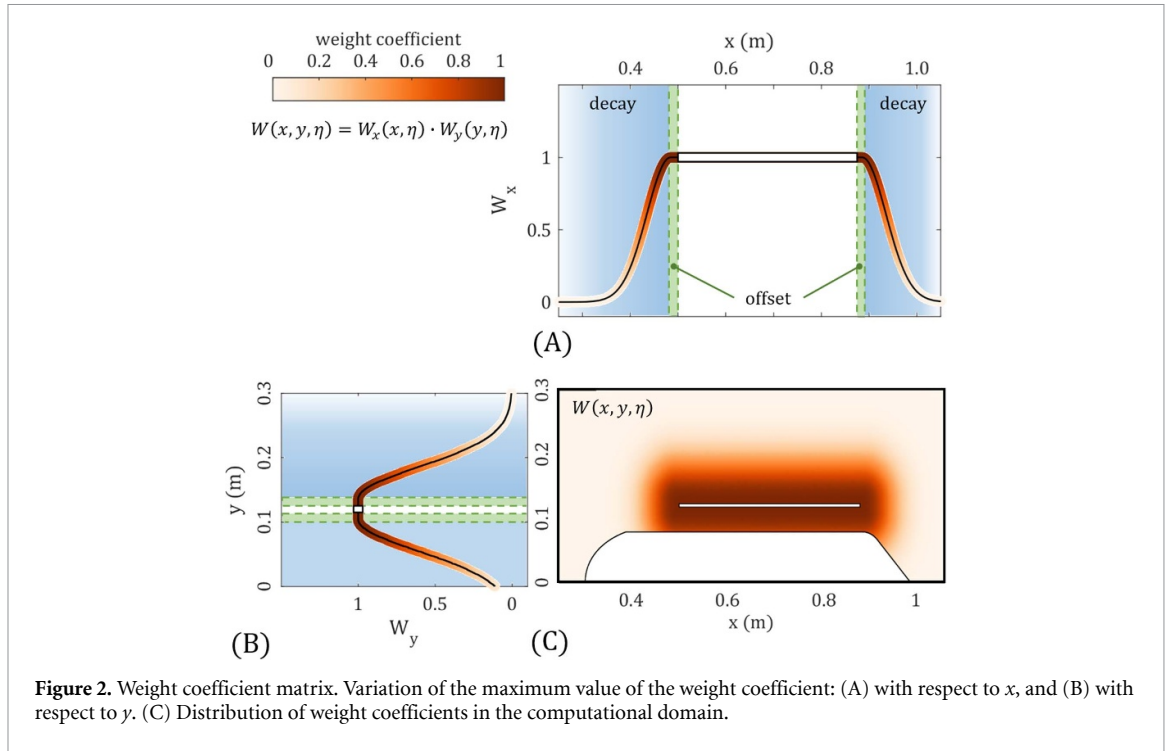


Figure 2. Weight coefficient matrix. Variation of the maximum value of the weight coefficient: (A) with respect to x , and (B) with respect to y . (C) Distribution of weight coefficients in the computational domain.

region R). To address this issue and identify these features, which are directly linked to the hydrodynamic forces acting on the robot, the solution states within the offset domain of the weight coefficient matrix (where the weight coefficient is 1) are extracted from R . Additionally, to ensure spatial continuity of the solution states, a decaying domain is defined. The weighted POD algorithm is employed to identify local features of the solution states in R . The weight coefficient matrix W is used to define rules for feature extraction. It is expressed as follows,

$$W(x, y, \eta) = W_x(x, \eta) \cdot W_y(y, \eta), \quad (8)$$

$$W_x(x, \eta) = \begin{cases} \exp\left(-\frac{(x-\mu_{x1})^2}{2\eta^2}\right) & x \in [0, \mu_{x1}] \\ 1 & x \in (\mu_{x1}, \mu_{x2}) \\ \exp\left(-\frac{(x-\mu_{x2})^2}{2\eta^2}\right) & x \in [\mu_{x2}, 2m] \end{cases}, \quad (9)$$

$$W_y(y, \eta) = \begin{cases} \exp\left(-\frac{(y-\mu_{y1})^2}{2\eta^2}\right) & y \in [0, \mu_{y1}] \\ 1 & y \in (\mu_{y1}, \mu_{y2}) \\ \exp\left(-\frac{(y-\mu_{y2})^2}{2\eta^2}\right) & y \in [\mu_{y2}, 0.5m] \end{cases}, \quad (10)$$

where $[\mu_{x1}, \mu_{x2}] \times [\mu_{y1}, \mu_{y2}] = [0.49, 0.89] \times [0.11, 0.13]$ (in metre) is the offset domain of the weight coefficient matrix. As depicted in figure 2, the weight coefficient decays in the computational domain as a Gaussian function.

To identify the fluid feature from the residual $R_{s_i}(r^i, \theta)$ with a satisfactory convergence rate (comparable to global feature identification, as depicted in figure 5), the snapshot POD analysis is

performed on the weighted residual $R_{s_i}^W(r_1^i, \eta, \theta) = W(\eta) \cdot R_{s_i}(r_1^i, \theta)$,

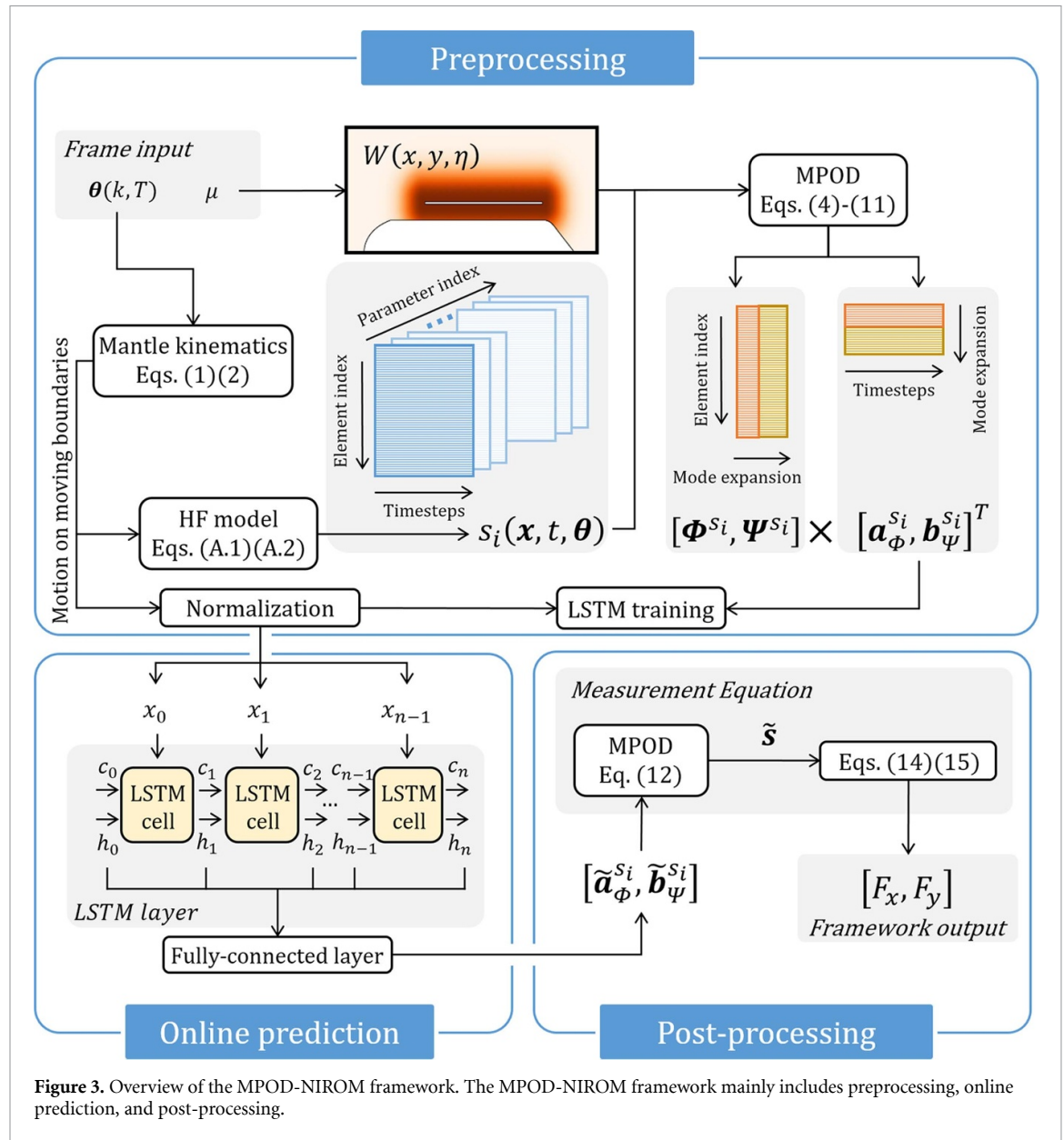
$$R_{s_i}^W(r_1^i, \eta, \theta) = \sum_{l=1}^{r_2^i} \Psi_l^{s_i}(x) b_l^{s_i}(t, \theta) + R_{s_i}^{2nd}(r_2^i, \eta, \theta), \quad (11)$$

where $\Psi_l^{s_i}(x)$ denotes the l th POD spatial mode required for the local feature identification, $b_l^{s_i}(t, \theta)$ denotes the temporal coefficient corresponding to $\Psi_l^{s_i}(x)$ and $R_{s_i}^{2nd}(r_2^i, \theta)$ denotes the residual after the local feature identification. By substituting equations (5) and (11) into equation (4), the solution states $s_i(x, t, \theta)$ are approximated by a linear combination of r_1^i set of POD modes defined in the global feature identification progress (represented by Φ^{s_i} and a^{s_i}) and r_2^i sets of POD modes defined in the local feature identification progress (represented by Ψ^{s_i} and b^{s_i}),

$$s_i(x, t, \theta) = \langle s_i \rangle + \sum_{j=1}^{r_1^i} \Phi_j^{s_i}(x) a_j^{s_i}(t, \theta) + \sum_{l=1}^{r_2^i} \Psi_l^{s_i}(x) b_l^{s_i}(t, \theta) + R_{s_i}^{2nd}(r_2^i, \eta, \theta). \quad (12)$$

2.3. Temporal evolution prediction of solution states

The temporal evolution of the solution states in the MPOD-based ROM is described by temporal coefficients $[(a_i^{s_i})^T (b_l^{s_i})^T]^T$ corresponding to the POD spatial modes $[\Phi_i^{s_i} \Psi_l^{s_i}]$. Considering the orthogonality of the global energy-favoured spatial modes (i.e. both $(\Phi_i^{s_i}, \Phi_j^{s_i}) = \delta_{ij}$) and the orthogonality of the residual



R_{s_i} to Φ^{s_i} (i.e. $(\Phi^{s_i}, R_{s_i}) = 0$), the evolution of the temporal coefficient $a_i^{s_i}(t, \theta)$ corresponding to $\Phi^{s_i}(x)$ can be obtained by substituting the decomposition in equation (5) into equation (4) [38]. Notably, as part of the residual R_{s_i} , the weighted residual $R_{s_i}^W$ is not orthogonal to Φ^{s_i} . Furthermore, the POD spatial mode extracted from the weighted residual Ψ^{s_i} is not orthogonal to Φ^{s_i} (i.e. $(\Phi^{s_i}, \Psi^{s_i}) \neq 0$). For this consideration, the evolution of the temporal coefficient $b_i^{s_i}(t, \theta)$ corresponding to $\Psi^{s_i}(x)$ cannot be obtained by the projection of the solution states onto Ψ^{s_i} . Therefore, a non-intrusive model based on ML is employed to predict the evolution of the POD temporal coefficients $\begin{bmatrix} (a_i^{s_i})^T & (b_i^{s_i})^T \end{bmatrix}^T$.

The proposed framework employs LSTM NN to construct a non-invasive model, owing to its potential capacity for learning and forecasting the temporal progression of sequences. By training a non-autoregressive LSTM NN, the mapping from the time

series of mantle motion to the temporal evolution of solution states (indicated by the time series of POD temporal coefficients) is acquired, which has the same inputs and outputs with the HF model. Precisely, POD temporal coefficients at the time step t_j are predicted by the trained LSTM NN (denoted by f_{LSTM}) based on the motion of the undulating mantle $q_m(t_j, \theta)$,

$$\begin{bmatrix} \tilde{a}^{s_i}(t_j, \theta) \\ \tilde{b}^{s_i}(t_j, \theta) \end{bmatrix} = f_{\text{LSTM}}(q_m(t_j, \theta)) \quad j = 1, 2, \dots, n_s, \quad (13)$$

where the predictions of temporal coefficients are denoted by $\tilde{a}^{s_i}$ and $\tilde{b}^{s_i}$. The proposed online prediction procedure, illustrated in figure 3, comprises an LSTM NN and a fully connected output layer. The current LSTM NN is trained using the adaptive moment estimation [64], a stochastic gradient

descent algorithm [65]. The hold-out method is employed to reorganise the dataset which involves extracting 20% of the total cases as test sets and dividing the remaining into training sets. The hyper-parameters for all training procedures are established: the learning rate is set to 0.001, and the batch size is 30. The NNs utilised in the current framework (detailed in appendix B) are based on the Deep Learning Toolbox in MATLAB.

The procedure for the developed MPOD-NIROM framework is summarised as follows.

Pre-processing:

- (i) Simulate the solution states of the HF model (equations (A.1) and (A.2));
- (ii) compute Φ^{s_i} , a^{s_i} , and R_{s_i} based on the global feature identification (equations (4) and (5));
- (iii) compute the weighted residual $R_{s_i}^W$ (equations (8)–(10));
- (iv) compute Ψ^{s_i} , b^{s_i} based on the local feature identification (equation (11));
- (v) train the LSTM NN to learn the mapping from input to output based on the adaptive moment estimation algorithm.

Online prediction:

- (i) Update the network input q_m by solving the parametric model of mantle kinematics (equations (1) and (2));
- (ii) predict the POD temporal coefficients (equation (13)).

Post-processing:

- (i) Reconstruct the predicted solution states based on the MPOD algorithm (equation (12));
- (ii) calculate the hydrodynamic forces working on the undulating mantle of the MUPRO (equations (14) and (15)).

As illustrated in figure 3, this paper proposes a framework that fully uses the proposed MPOD algorithm and LSTM-based NIROM to predict the hydrodynamic force working on the underwater robot (denoted by $F = [F_x, F_y]$). F is calculated from the prediction of the solution states as follows,

$$F_x = \int_{\Gamma_m} (-pn_x + 2\mu u_x n_x + \mu(u_y + v_x) n_y) d\Gamma_m, \quad (14)$$

$$F_y = \int_{\Gamma_m} (-pn_x + \mu(v_x + u_y) n_x + 2\mu v_y n_y) d\Gamma_m. \quad (15)$$

3. Results and discussion

3.1. Validation research for the HF model

In this section, the mesh independence of the HF model is verified. The computational domain is

partitioned based on three distinct grid strategies (i.e. 28 079 elements with the minimum unit size 0.003 m, 38 552 elements with the minimum unit size 0.002 m, and 45 186 elements with the minimum unit size 0.001 m). Table 1 illustrates the time-averaged value of the instantaneous hydrodynamic force acting on the mantle in the negative x -axis direction $\langle F_x \rangle$ (F_x is defined in equation (A.4)). Since the relative change between T2 and T3 is minimal (only 1.2% when $\theta_{C1} = [0.5, 2]$), the mesh with 38 552 elements is used for further simulations. Additionally, we discuss simulations of the HF model with three different time steps. Table 1 also shows $\langle F_x \rangle$ simulated by the HF model with 38 552 elements for three different time steps (i.e. $dt = 0.0006$ s, $dt = 0.0005$ s, $dt = 0.0004$ s). It can be observed that the HF model with a time step of 0.0005 s accurately predicts the hydrodynamic forces acting on the mantle, as there are no significant changes between T2 and T5 (only 0.2% when $\theta_{C2} = [0.4, 2]$).

Figure 4(A) illustrates the instantaneous hydrodynamic force F_x for the cases (T1, T2 and T3) with different mesh sizes when $\theta_{C2} = [0.4, 2]$. Between $t = 3T$ and $t = 4T$, the peak of F_x occurs near $t = 3.71T$ (the peak value of F_x is 5.8390 N when the number of elements is 45 186, the peak value of F_x is 5.8184 N when the number of elements is 38 552, and the peak value of F_x is 5.7329 N when the number of elements is 28 079). The relative changes in peak hydrodynamic forces between T2 and T3 and between T1 and T2 are 0.3% and 1.5%, respectively. F_x at different time steps are shown in figure 4(B) (T4, T2 and T5). The peak value of F_x is 5.8229 N when $dt = 0.0006$ s, 5.5184 N when $dt = 0.0005$ s, and 5.7732 N when $dt = 0.0004$ s. Simultaneously, the relative changes of the peak value of F_x between T5 and T2 as well as between T2 and T4 are 0.08% and 0.8%, respectively. Therefore, considering the computational cost and reliability, the solution states presented in this paper are all based on T2 with a mesh division of 38 522 elements and a time step of $dt = 0.0005$ s.

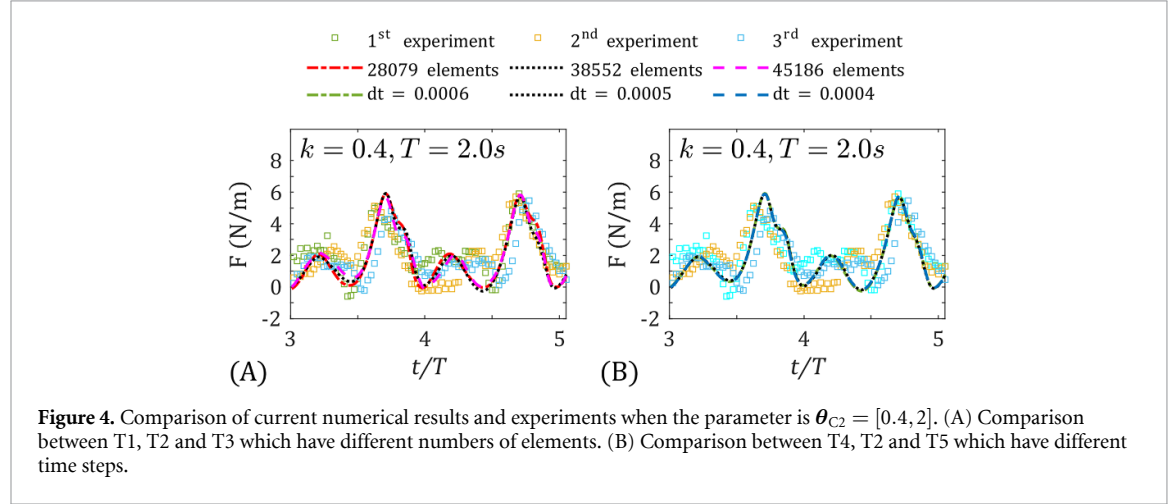
The HF model's validity is ascertained through a comparative analysis of $\langle F_x \rangle$, using three different parameter combinations ($\theta_{C1}, \theta_{C2}, \theta_{C3}$) between the HF model simulations and experimental measurements. The presented figures 10(A)–(C) illustrates a satisfactory and consistent trend. In each propulsion cycle, two peaks of varying magnitudes are observed, which aligns with the experiment of the oblate jellyfish-inspired robot experiment [22].

3.2. POD convergence results

Based on the HF model configuration described in the previous section, all 60 cases for various parameter combinations (i.e. $\theta(k_i, T_l)$) where the dimensionless wavenumber $k_i = 0.1 + 0.05i$, undulating period $T_l = (1.5 + 0.5l)$ s, $i = 0, 1, \dots, 14$ and $l = 0, 1, 2, 3$ are simulated. For each case, regularly spaced snapshots are generated during

Table 1. A sensitivity study on the number of elements and time steps, i.e. the value of $\langle F_x \rangle$ for three cases with different parameter combinations (θ_{C1} , θ_{C2} , and θ_{C3}).

Test Case	Number of Elements	dt/T	$\theta_{C1} = [0.5, 2]$		$\theta_{C2} = [0.4, 2]$		$\theta_{C3} = [0.2, 2]$	
			$\langle F_x \rangle$ in N	Changes	$\langle F_x \rangle$ in N	Changes	$\langle F_x \rangle$ in N	Changes
T1	28 079	0.000 25	1.5998	—	1.9085	—	1.9939	—
T2	38 552	0.000 25	1.5200	5.0%	1.9312	1.2%	1.9706	1.2%
T3	45 186	0.000 25	1.5016	1.2%	1.9535	0.6%	1.9698	0.4%
T4	38 552	0.000 30			2.0008	—		
T2	38 552	0.000 25			1.9312	0.4%		
T5	38 552	0.000 20			1.9264	0.2%		



two mantle undulation cycles (from $3T$ to $5T$) with a dimensionless time step $\Delta t/T = 0.05$. The total number of snapshots of the solution states is $n^{s_i} = [n^u, n^v, n^p] = [2460, 2460, 2460]$. The snapshot POD analysis is performed on the fluctuating component of the solution states s_i' (equations (4) and (5)). Convergence plots for RIC^{s_i} are shown in figure 5.

RIC^{s_i} Converges to 1 as the expansion of the index of the POD spatial mode. As shown in figures 5(A)–(C), there is a constant $c_n^{s_i} = [c_n^u, c_n^v, c_n^p]^T = [0.9965, 0.9965, 0.9964]^T$ establishes the following formula,

$$\lim_{r_1^{s_i} \rightarrow n^{s_i}} (1 - \text{RIC}^{s_i}(r_1^{s_i} + 1)) (1 - \text{RIC}^{s_i}(r_1^{s_i}))^{-1} = c_n^{s_i}. \quad (16)$$

Here, the RIC of the snapshot POD algorithm tends to converge sub-linearly when $r_1^{s_i}$ approaches n^{s_i} . This phenomenon is consistent with the previous studies [66, 67] and proves that the truncation error $\varepsilon^{s_i} = 1 - \text{RIC}^{s_i}$ and the expansion degree of the POD-ROM j (i.e. DoF of the POD spatial mode) satisfy the following equation,

$$\varepsilon^{s_i} = 1 - \text{RIC}^{s_i} \approx O\left(j^{-\frac{p_\varepsilon}{n^{s_i}}}\right), \quad (17)$$

where p_ε denotes the parameter to be determined.

On the other hand, as shown in figure 5, $r_1^{s_i} = [r_1^u, r_1^v, r_1^p] = [26, 47, 9]$ out of $n^{s_i} = 3075$ pairs of POD modes are retained for the ROM, accompanied by the critical convergence rate $c_m^{s_i} = [c_m^u, c_m^v, c_m^p] = [0.97, 0.97, 0.91]$, and $\text{RIC}^{s_i} = [\text{RIC}^u, \text{RIC}^v, \text{RIC}^p] = [0.930, 0.956, 0.959]$. Notably, only the first 9 POD modes of the pressure are retained since the relative information content $\text{RIC}^p(9)$ exceeded $\text{RIC}^u(29)$ or $\text{RIC}^v(47)$. As shown in table 2 (Case 1), if the convergence rate is controlled by $c_m^{s_i}$, the reliability of the classical POD-based ROM is unsatisfactory ($\text{NMSE}(s_i|_{\Gamma_m}) \leq 0.2\%$), especially for the pressure on the moving boundary $p|_{\Gamma_m}$.

3.3. MPOD convergence results

The primary novelty of the MPOD algorithm proposed in this manuscript is the fast convergence of the fluid feature near the moving boundary relating to the DoF, which is denoted by j . The efficacy of the fluid feature identified by the MPOD algorithm is evaluated utilising the projection error, expressed by the normalised mean squared error (NMSE, relative to the variance),

$$\text{NMSE}(s_i|_{\Gamma_m}) = \frac{1}{n_\theta} \sum_{l=1}^{n_\theta} \int_{\Gamma_m} (s_i^{\text{HF}}(\theta_l) - s_i^{\text{ROM}}(\theta_l))^2 \times d\Gamma_m \int_{\Gamma_m} (s_i^{\text{HF}}(\theta_l) - s_i^{\text{HF}}(\theta_l))^2 d\Gamma_m^{-1}. \quad (18)$$

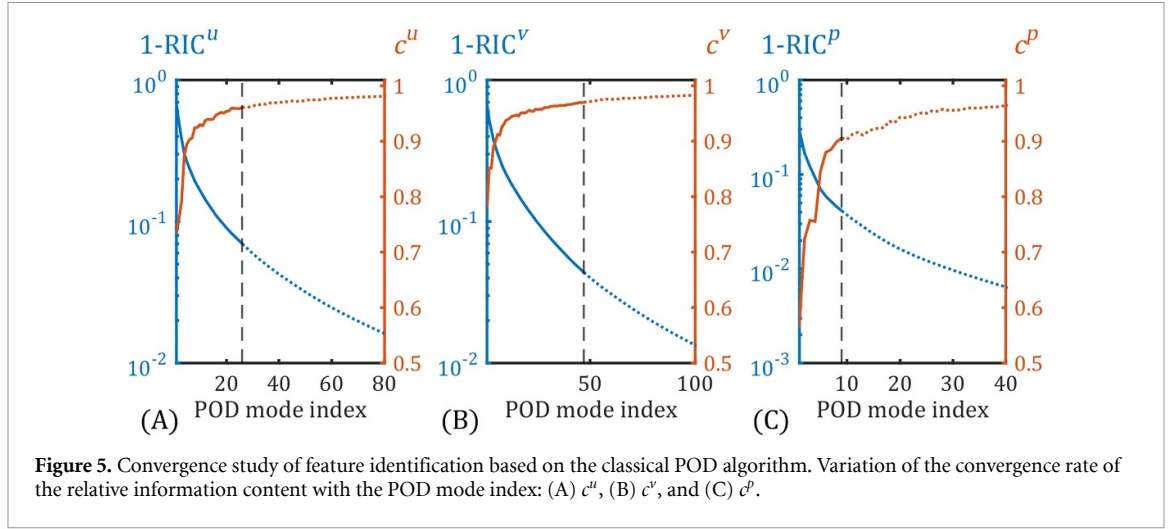


Figure 5. Convergence study of feature identification based on the classical POD algorithm. Variation of the convergence rate of the relative information content with the POD mode index: (A) c^u , (B) c^v , and (C) c^p .

Table 2. Comparison of L^2 errors for feature identification result between the classical POD algorithm and the proposed MPOD algorithm: $\text{NMSE}(s_i|_{\Gamma_m})$ and $\text{rMSE}(u_\tau)$ in different degrees of freedom r^{s_i} .

η	MPOD algorithm					POD algorithm		
	1	0.5	0.25	0.1	0.05	Case 1	Case 2	Case 3
r^u	180	139	113	98	95	26	98	223
$\text{NMSE}(u _{\Gamma_m})$	0.19%	0.20%	0.20%	0.20%	0.20%	0.82%	0.50%	0.20%
r^v	145	119	95	72	69	47	72	134
$\text{NMSE}(v _{\Gamma_m})$	0.20%	0.20%	0.20%	0.20%	0.20%	0.93%	0.55%	0.20%
r^p	87	63	48	34	33	9	34	87
$\text{NMSE}(p _{\Gamma_m})$	0.20%	0.20%	0.20%	0.20%	0.20%	6.12%	1.30%	0.20%
$\text{rMSE}(u_\tau)$	0.88%	0.90%	0.89%	0.89%	0.88%	3.18%	1.85%	0.90%

Considering that the non-slip condition is applied on the boundary Γ_m ,

$$u_\tau = u \cdot n_y + v \cdot n_x = 0, \quad (19)$$

where u_y denotes the tangential velocity on Γ_m (of which the actual value is 0), and $n = [n_x, n_y]$ is the normal unit vector of Γ_m . The mean squared error (MSE) of u_τ^{ROM} is used as an additional metric to estimate the accuracy of the ROM,

$$\text{MSE}(u_\tau^{\text{ROM}}) = \frac{1}{n_\theta} \sum_{l=1}^{n_\theta} \left(\left\langle \int_{\Gamma_m} (u_\tau^{\text{ROM}}(\theta_l) - 0)^2 d\Gamma_m \right\rangle \right). \quad (20)$$

For simplicity, $\text{MSE}(u_\tau^{\text{ROM}})$ is subsequently presented in relative increments, namely rMSE , which is the increase relative to the HF model defined as follows,

$$\text{rMSE}(u_\tau) = 1 - \text{MSE}(u_\tau^{\text{ROM}}) \text{MSE}(u_\tau^{\text{HF}})^{-1}. \quad (21)$$

To reliably predict the fluid feature near the moving boundary Γ_m , local feature identification is employed in the proposed algorithm where the index expansion of Ψ^{s_i} will progress until $\text{NMSE}(s_i|_{\Gamma_m}) \leq 0.2\%$.

The Gaussian attenuation coefficient η determines the attenuation region size, which is crucial for in local feature identification. Hence, an analysis on η for all the cases is conducted in the present study. The parameter η for the weight coefficient matrix $W(\eta)$ is tabulated in table 2, along with the DoF of MPOD-based ROM $r^{s_i} = r_1^{s_i} + r_2^{s_i}$ ($r_1^{s_i}$ refers to the number of Φ^{s_i} introduced by the global feature identification process and $r_2^{s_i}$ indicates the number of Ψ^{s_i} employed by the local feature identification process) and metrics to characterise the quality of the ROM near the moving boundary (i.e. $\text{NMSE}(s_i|_{\Gamma_m})$ and $\text{rMSE}(u_\tau)$). When $\eta = 0.1$, halving the η did not significantly reduce the DoF required by the proposed model (halving the η can reduce the number of DoF = $r^u + r^v + r^p$ by about 4%, i.e. from 204 to 197). Hence, subsequent results, based on the proposed MPOD algorithm presented later is based on $\eta = 0.1$.

The proposed MPOD algorithm is designed to predict the propulsion force acting on the robot's moving boundary. For comparison, two cases based on the classical POD algorithm (case 2 and case 3) are shown in table 2. The minimum change in $\text{NMSE}(s_i|_{\Gamma_m})$ between the proposed MPOD algorithm and Case 2 is 40% when the DoF are the same (where r^u , r^v and r^p are the DoF values of the MPOD algorithm). Comparing the proposed MPOD algorithm and Case 3, when $\text{NMSE}(s_i|_{\Gamma_m})$ is the

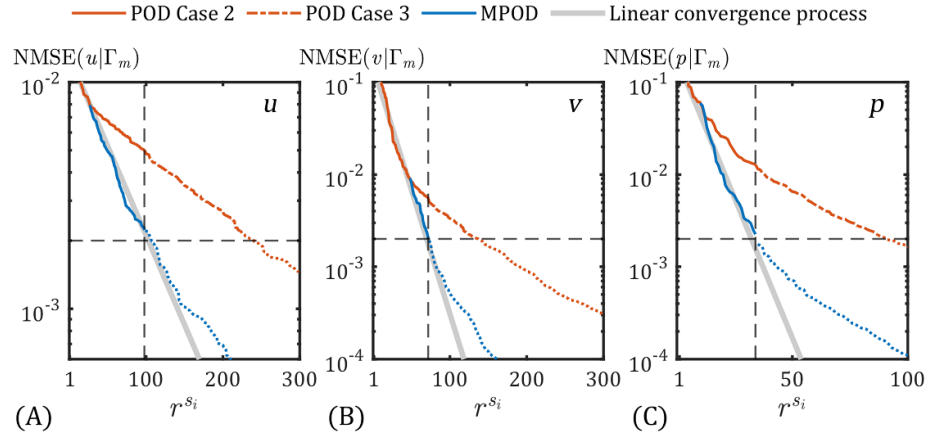


Figure 6. Variations of L^2 errors with the different degrees of freedom r^s : (A) $\text{NMSE}(u|\Gamma_m)$, (B) $\text{NMSE}(v|\Gamma_m)$, and (C) $\text{NMSE}(p|\Gamma_m)$. The orange line indicates the results of the classical POD algorithm: the solid and dashed orange lines are for Case 2 and Case 3, respectively. The blue line indicates the result of the proposed MPOD algorithm, i.e. the solid blue line represents the result of MPOD with $\eta = 0.1$ and $\text{NMSE}(s_i|\Gamma_m) \leq 0.2\%$. The grey line shows that the error reduction follows linear convergence.

same, the total DoF of the proposed MPOD algorithm is 46% that of the classical POD algorithm. These results show that the proposed MPOD algorithm efficiently identifies fluid features near the boundary.

For intuition, the convergence plots for $\text{NMSE}(s_i|\Gamma_m)$ are shown in figure 6, wherein the $\text{NMSE}(s_i|\Gamma_m)$ of the proposed MPOD algorithm converges faster than the classical POD algorithm. Furthermore, the $\text{NMSE}(s_i|\Gamma_m)$ of the proposed MPOD algorithm converges almost linearly when $r^u \leq 98$, $r^v \leq 72$, or $r^p \leq 34$. The linear convergence processes for benchmarking are denoted as solid grey lines in figure 6, with the approximate linear convergence rate $c_{\text{lin}}^s = [c_{\text{lin}}^u, c_{\text{lin}}^v, c_{\text{lin}}^p] = [0.98, 0.94, 0.87]$. Furthermore, the feature identification based on the proposed MPOD algorithm (described by the blue line in figure 6) is more efficient than that based on the classical POD algorithm (represented by the orange line in figure 6).

3.4. Validation research for online prediction

The proposed framework employs LSTM-based NIROM for online prediction of POD temporal coefficients. This section examines the suitability of the current NIROM for predicting transient hydrodynamics by assessing error estimation and computational cost.

The current NIROM employs the half-mean-square error (HMSE) as the loss function for error estimation, which is defined as follows,

$$\text{HMSE} = \frac{1}{2n_s} \sum_i^{n_s} \left(\sum_j^r \text{std}(a_j(t_i) - \tilde{a}_j(t_i))^2 + \sum_j^{r_2} \text{std}(b_j(t_i) - \tilde{b}_j(t_i))^2 \right), \quad (22)$$

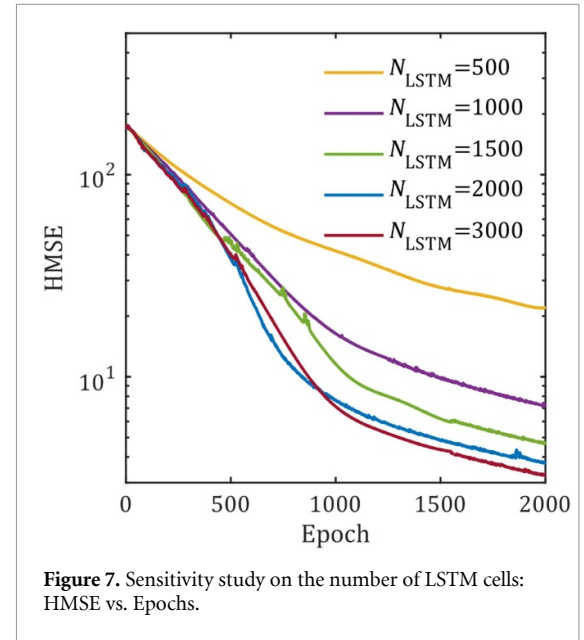
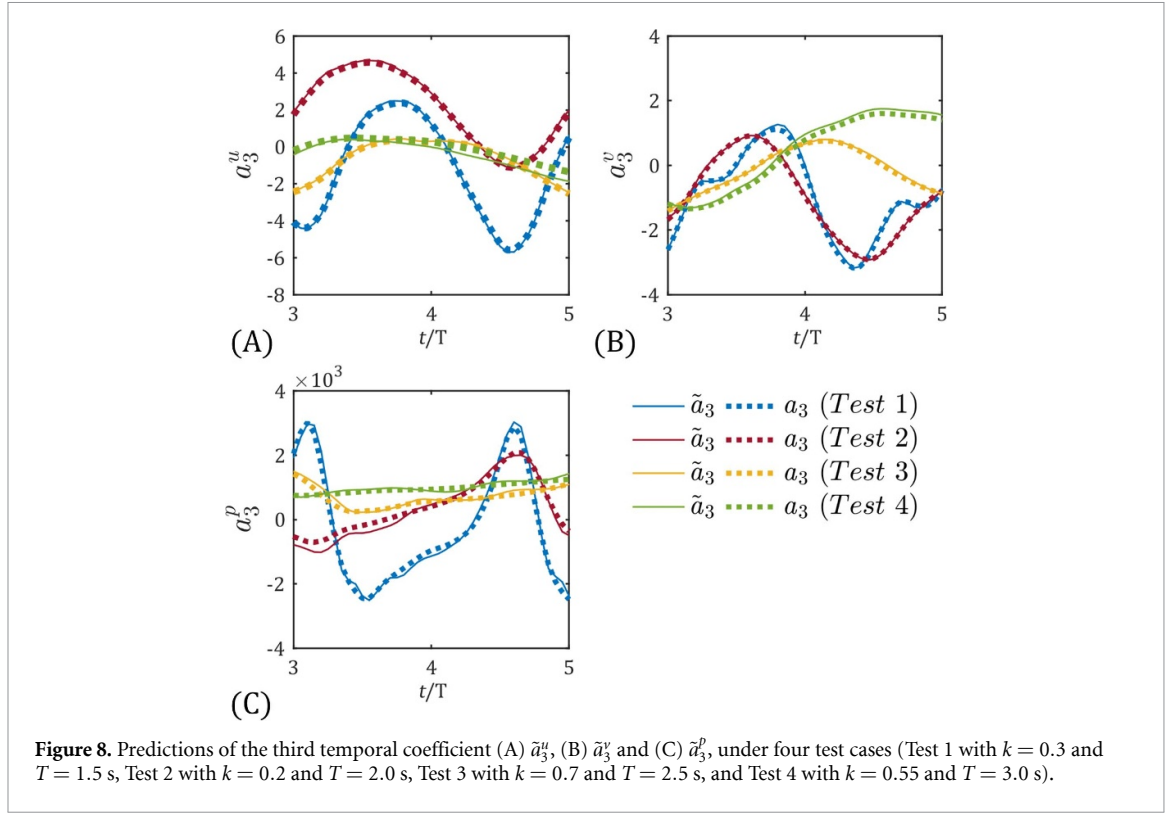


Figure 7. Sensitivity study on the number of LSTM cells: HMSE vs. Epochs.

where $\text{std}(a_i - \tilde{a}_i) = ((a_i - \tilde{a}_i) - a_i) \cdot \sigma(a_i)^{-1}$ denotes the mean-variance normalisation. In this work, temporal coefficients are mean-variance normalised because the order of magnitude of the temporal coefficient differs in the fluid velocity and the pressure. For time-series data predicting, the number of LSTM cells impacts the accuracy of the LSTM-based NIROM. The prediction errors of the LSTM-based NIROMs with different numbers of LSTM cells N_{LSTM} are shown in figure 7. It is observed that the percentage variation in HMSE between $N_{\text{LSTM}} = 1500$ and $N_{\text{LSTM}} = 2000$ is 26.1%, while the percent change in HMSE between $N_{\text{LSTM}} = 2000$ and $N_{\text{LSTM}} = 3000$ is 14.9% at the last epoch. Considering the computational cost and prediction reliability, the number of LSTM cells of the current NIROM is $N_{\text{LSTM}} = 2000$.



As mentioned in section 2.3, 20% of the dataset (12 cases) is used as the testing data, figure 8 illustrates the prediction (represented by the solid line) for the third temporal coefficient $\tilde{a}_3^u$, $\tilde{a}_3^v$ and $\tilde{a}_3^p$ (corresponding to the third POD spatial modes Φ_3^u , Φ_3^v and Φ_3^p , respectively) for the four cases (i.e. Test 1 with $k = 0.3$ and $T = 1.5$ s, Test 2 with $k = 0.2$ and $T = 2.0$ s, Test 3 with $k = 0.7$ and $T = 2.5$ s, and Test 4 with $k = 0.55$ and $T = 3.0$ s) within the independent test set. The predictions for these test cases show that the current LSTM-based NIROM predictions $\tilde{a}$ have a consistent trend with the reference values a . Simultaneously, the coefficient of determination R^2 for the test sets is $R^2(a^u) = 0.9278$, $R^2(a^v) = 0.9422$ and $R^2(a^p) = 0.9101$, (the average of all 12 test cases), R^2 is defined as follows,

$$R^2(a) = 1 - \frac{\sum_{j=1}^r \sum_{i=1}^{n_t} (a_j(t_i) - \tilde{a}_j(t_i))^2}{\sum_{j=1}^r \sum_{i=1}^{n_t} (a_j(t_i) - \bar{a}_j)^2}. \quad (23)$$

Thus, the proposed LSTM-based NIROM can reliably provide predictions for cases independent of the training set (but with the parameters restricted to $k \in [0.1, 0.8]$ and $T \in [1.5, 3]$).

As described in equation (13), for the cases discussed in this paper, the POD temporal coefficients are predicted by the employed LSTM-based NIROM (represented by $\left[(\tilde{a}_i^{s_i})^T (\tilde{b}_i^{s_i})^T \right]^T$), taking the undulating motion of the robot (represented by q_m defined in equation (A.1)) as input. Substituting $\tilde{a}_i^{s_i}$ and

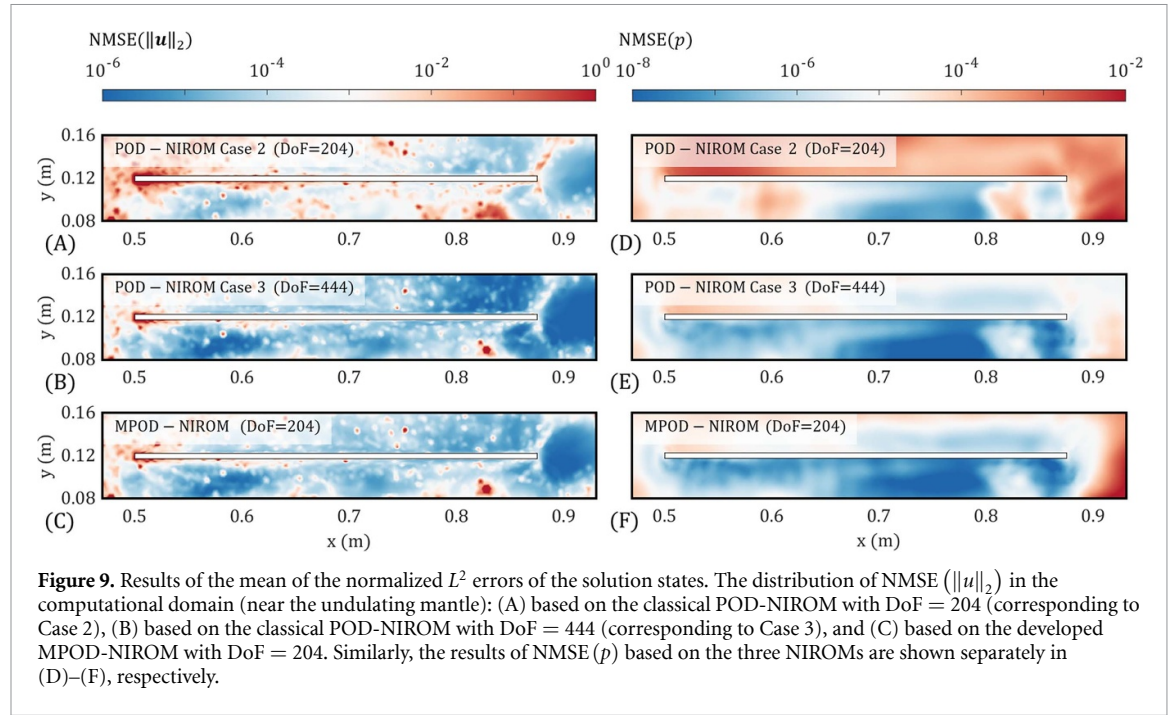
$\tilde{b}_i^{s_i}$ into equation (12), the prediction of the solution states $\tilde{s}_i^{\text{ROM}}$ were obtained. The discrepancies in predicting the solution states are compared between the developed MPOD-NIROM framework and the classical POD-NIROM (i.e. case 2 and case 3). The NMSE of the solution state at each grid point in the computational domain between the ROM prediction and the HF model simulation is defined as follows,

$$\text{NMSE}(s_i) = \frac{1}{n_\theta} \sum_l \left\langle \left(\tilde{s}_i^{\text{ROM}}(\theta_l) - s_i^{\text{HF}}(\theta_l) \right)^2 \times \left(s_i^{\text{HF}}(\theta_l) - s_i^{\text{HF}}(\theta_j) \right)^2 \right\rangle^{-1}. \quad (24)$$

The NMSE of the 2-norm of the fluid velocity $\text{NMSE}(\|u\|_2)$ near the motion boundary is shown in figures 9(A)–(C), and $\text{NMSE}(p)$ is shown in figures 9(D)–(F). In figure 9, the colour intensity is linearly related to $\log_{10}(\text{NMSE}(s_i))$.

Figure 9(C) and (E) demonstrate the reliability of predictions obtained using the developed MPOD-NIROM. In all 60 cases examined in this paper, the NMSE of the proposed model is satisfactory around the moving boundary, indicating the reliability of the developed framework.

Comparing the results presented in figures 9(A) and (D) for the classical POD-NIROM (Case 2), it can be observed that the NMSE of the developed MPOD-NIROM is of a lower order of magnitude when the DoF is the same. Furthermore, the NMSE of the developed MPOD-NIROM is similar to the classical POD-NIROM (Case 3), as shown in figures 9(B)



and (E), when the grid points are located near the moving boundary.

These findings suggest that the developed MPOD-NIROM framework can reliably predict the temporal evolution of local fluid features at grid points in the computational domain near the moving boundary). However, it should be noted that the order of magnitude of the NMSE in the predictions of the proposed framework is higher than that of the POD-NIROM (Case 3) when the grid points are far away from the moving boundary. This may be attributed to the weighting of the residual term R (as shown in figure 2) when identifying local features. Thus, fluid features at grid points far from Γ_m could not be fully identified.

3.5. Prediction of the hydrodynamic force

To visually demonstrate the effectiveness of mantle undulated propulsion and the accuracy of hydrodynamics force prediction acting on the robot, figures 10(A)–(C) presents the temporal evolution of F_x over two swimming cycles (from $t = 3T$ to $t = 5T$). Both simulations (using the HF model) and predictions (using ROMs) are consistent with the experimental measurements. The results indicate two propulsion peaks in each swimming cycle, with the primary stroke producing the most pronounced peak and the sub-stroke producing the second most significant peak. This phenomenon is similar to fish propulsion research (referring to forward and backward tail strokes) [5, 68, 69]. However, the magnitudes of the two propulsion peaks are different due to the different conditions on both sides of the moving boundary Γ_m . One side of Γ_m is close to the wall, whereas the other is open water, causing the discrepancy. Specifically, when the propulsion force

is at its most significant peak, the mantle's end is closest to the wall. In contrast, when the propulsion force reaches the second most prominent peak, the mantle's end is furthest from the wall. This occurrence aligns with existing experimental research on jellyfish-inspired robots [20, 70]. It is noteworthy that during the sub-stroke, the hydrodynamic force acting on the MUPRO still provides propulsion. In contrast, certain jellyfish-inspired robots may not provide propulsion or generate resistance at this time.

For intuition, the normalised squared error (NSE) of the prediction of $F_x(t; \theta_l)$ is calculated as follows,

$$NSE(F_x, t; \theta_l) = \frac{(F_x^{HF}(t; \theta_l) - F_x^{ROM}(t; \theta_l))^2}{\langle F_x^{HF}(t; \theta_l) - \langle F_x^{HF}(\theta_l) \rangle \rangle^2}, \quad (25)$$

which is shown in figures 10(D)–(F) as scatter points, and $\langle NSE(F_x, t; \theta_l) \rangle$ is shown as the horizontal line. Here, F_x is well predicted by the developed MPOD-NIROM framework with $DoF = 204$. The NSE of the proposed framework is similar to that of the classical POD-NIROM with $DoF = 444$ (Case 3), whereas those of the classical POD-NIROM with $DoF = 204$ (Case 2) are inaccurate for the three cases presented in figure 10. Further, the NMSE of the prediction of F_x for all 60 cases employed in this paper is calculated as follows,

$$NMSE(F_x, \theta_l) = \frac{\langle (F_x^{HF}(\theta_l) - F_x^{ROM}(\theta_l))^2 \rangle}{\langle (F_x^{HF}(\theta_l) - \langle F_x^{HF}(\theta_l) \rangle)^2 \rangle^{-1}}. \quad (26)$$

Figure 11 shows the $NMSE(F_x, \theta_l)$ of the 60 cases, where the colour indicates the $NMSE(F_x)$

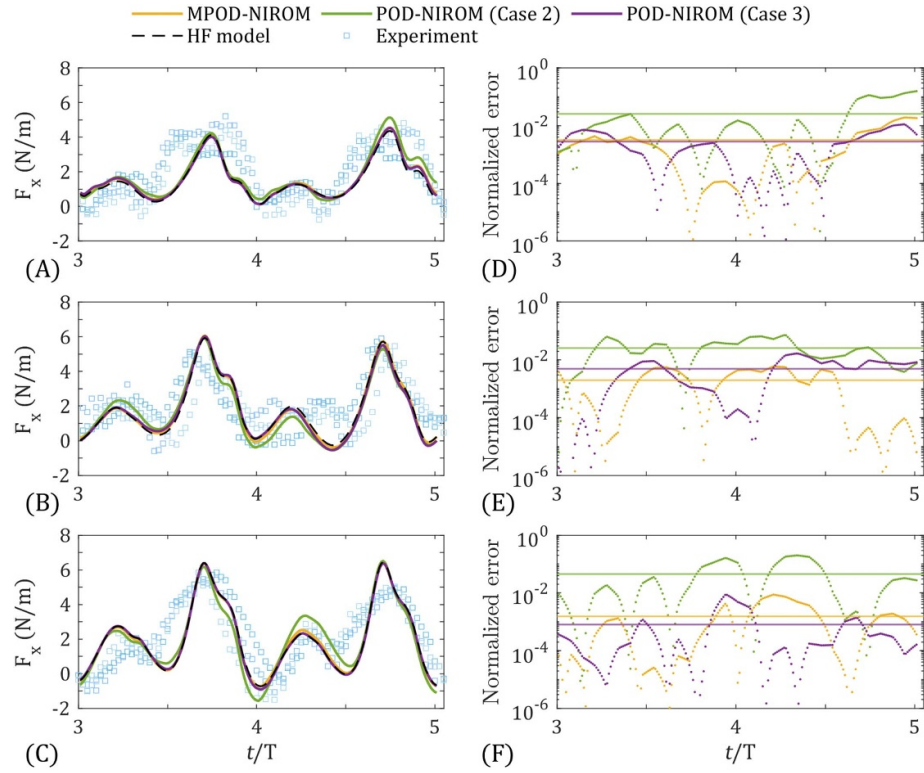


Figure 10. Results of NIROMs in predicting the hydrodynamic forces working on the undulating mantle: time histories of F_x and corresponding NSE for three cases, i.e. (A) (D) for the case with $\theta_{C1} = [0.5, 2]$, (B) (E) for the case with $\theta_{C2} = [0.4, 2]$, and (C) (F) for the case with $\theta_{C3} = [0.2, 2]$. In (D)–(F), the horizontal line represents the time average of NSE between $t = 3T$ and $t = 5T$.

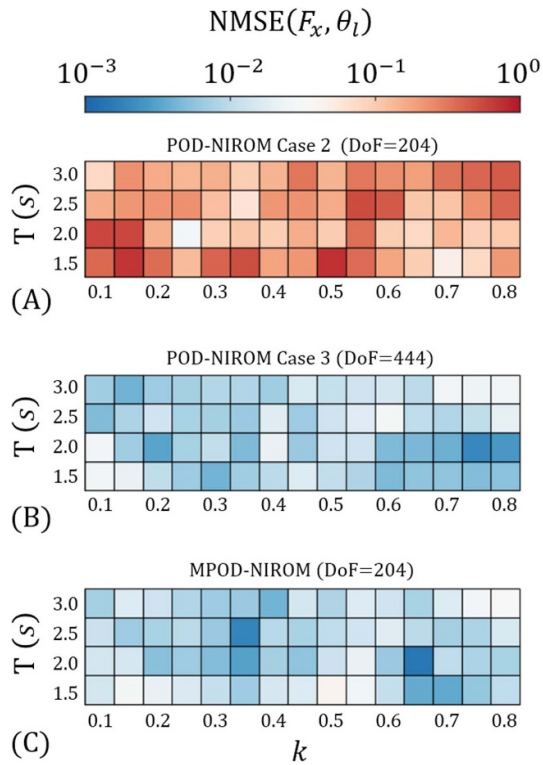


Figure 11. Result of the prediction for all cases, i.e. $NMSE(F_x, \theta_l)$ based on: (A) the classical POD-NIROM with $DoF = 204$ (corresponding to Case 2), (B) the classical POD-NIROM with $DoF = 444$ (corresponding to Case 3), and (C) the developed MPOD-NIROM framework with $DoF = 204$.

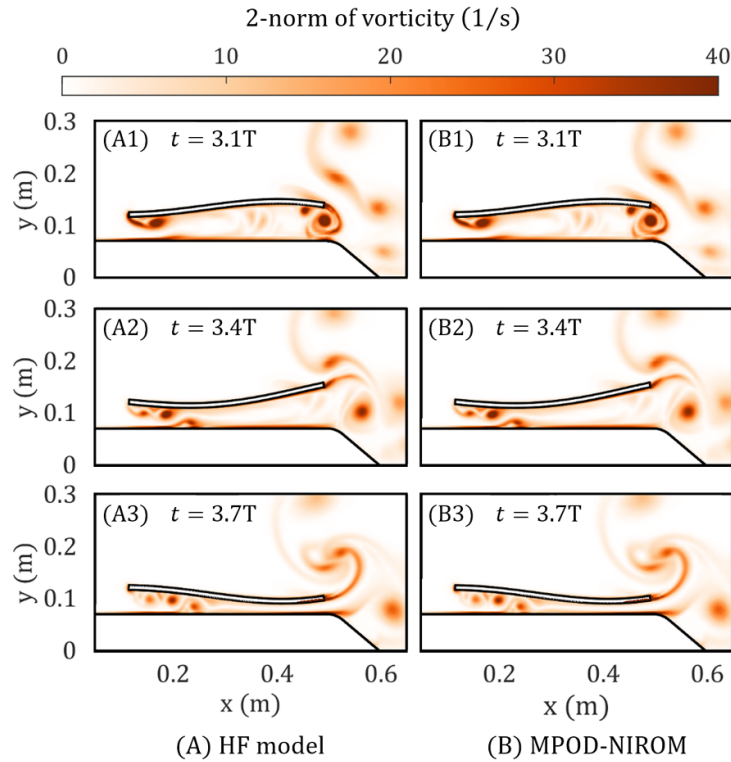


Figure 12. Instantaneous 2-norm of vorticity. (A) The HF model. (B) The developed MPOD-NIROM framework with DoF = 204.

of the corresponding case θ_l . For comparison, the average NMSE(F_x, θ_l) of the cases is calculated, yielding 1.09×10^{-1} for the classical POD-NIROM with DoF = 204, 4.9×10^{-3} for the classical POD-NIROM DoF = 444, and 5.2×10^{-3} for the developed MPOD-NIROM with DoF = 204. Intuitively, for predicting the hydrodynamic forces (F_x) acting on the undulating mantle Γ_m , the developed MPOD-NIROM requires fewer DoF than the classical POD-NIROM to obtain a similar NMSE(F_x, θ_l). Moreover, the NMSE(F_x, θ_l) of the proposed model is 1 or 2 orders of magnitude less than that of the classical model with the same DoF. The results presented herein demonstrate the capability of the developed MPOD-NIROM framework to predict the temporal evolution of local fluid features, particularly the hydrodynamic forces exerted on a moving boundary. However, the applicability of the proposed framework should be validated using discussions on other aquatic-inspired robots.

3.6. Discussions on wake structure

In this study, we developed a 2D simulation data-driven NIROM based on the proposed MPOD algorithm, aiming to accurately and quickly predict the hydrodynamic force acting on the biomimetic underwater robot (MUPRo). To discuss the predictive performance of the proposed MPOD-NIROM, the vorticity surrounding the robot at typical moments from 3T to 4T are shown in figure 12. When $t = 3T$, the trailing edge of the mantle starts to expand away

from the main cabin, forming a vortex ring in the channel between the mantle and the main cabin (as shown in the first row of figure 12). Subsequently, the vortex ring begins to move downstream and the hydrodynamic force present the robot a propulsion. After $t = 3.5T$, the mantle begins to contract towards the main cabin, resulting in a vortex ring on the outside of the trailing edge of the mantle. This vortex ring moves downstream together with the vortex ring formed previously. (as shown in the third row of figure 12). By comparing figures 12(A) and (B), we find that the predictions based on the MPOD-NIROM and the HF model exhibit the same trend. Similar tail vortex evolution processes are found in existing experimental studies [10, 11].

3.7. Model-based offline parameter planning

The proposed MPOD-NIROM significantly reduces the computational cost of hydrodynamic simulation for underwater robots to the millisecond level, enabling offline parameter planning for Model-Based offline parameter planning of MUPRo to be completed within a shorter time period. The offline parameter planning problem of MUPRo is solved within a 2D parameter plane, where $k \in [0.1, 0.8]$ with a step size of $\Delta k = 0.01$, and $T \in [1.5, 3.0]$ with a step size of $\Delta T = 0.03$ s. The optimization objective is to determine the control parameters starting from the specified initial stage ($\theta_{s10} = [0.5, 3]$ in Task 1, $\theta_{s20} = [0.7, 3]$ in Task 2), aiming to maximize $\langle F_x \rangle$. The magnitude of

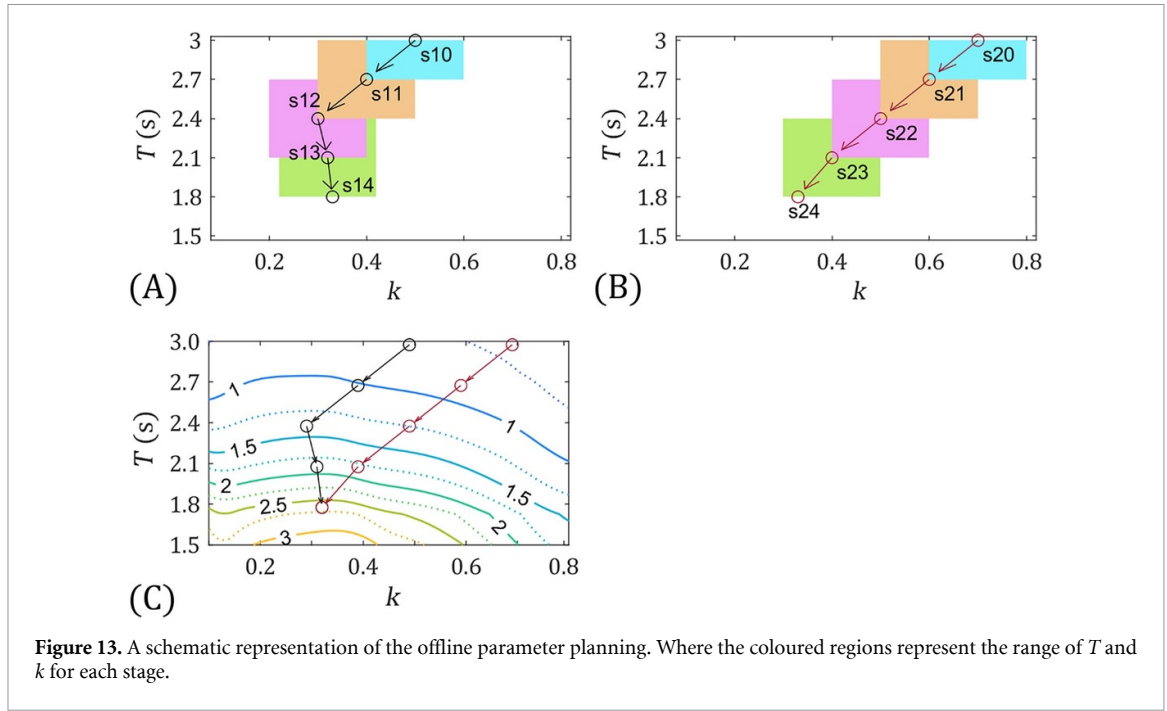


Table 3. Results of offline parameter planning (Task 1) based on MPOD-NIROM and experimental validations based on the 2D prototype. All calculations are implemented on a computer with a 12-core CPU (i7 12 700), 12G dedicated memory GPU (RTX 3080 Ti), and 64G DDR4 running memory.

Stage	s0	s1	s2	s3	s4
k	0.5	0.4	0.3	0.32	0.33
T	3	2.7	2.4	2.1	1.8
$\langle F_x \rangle^{\text{MPOD}} (\text{Nm}^{-1})$	0.799	0.987	1.353	1.833	2.578
$\langle F_x \rangle^{\text{exm}} (\text{Nm}^{-1})$ in 1st undulation period	0.734 (−8.1%)	0.935 (−5.3%)	1.334 (−1.4%)	1.891 (3.2%)	2.636 (2.2%)
$\langle F_x \rangle^{\text{exm}} (\text{Nm}^{-1})$ in 2nd undulation period	0.753 (−5.7%)	0.994 (7.1%)	1.318 (−2.6%)	1.903 (3.8%)	2.707 (5.0%)
^a Computational cost (ms)	9533	144 74	143 19	143 55	106 48
Number of cases calculated	200	300	300	300	220

^a All the parallel computing process is based on 12 workers.

parameter variations in each stage is constrained to $\pm 10\Delta k$ and $\pm 10\Delta T$. Let $\langle F_x(\theta_i) \rangle$ represent $\langle F_x \rangle$ in the i th stage, where we need to search for the maximum average increment in hydrodynamic force $\max\langle \Delta F_x(\theta_{i-1}) \rangle$ caused by parameter variations within the feasible range of the i th stage. Based on the principle of Bellman optimality [71], this optimization problem can be described using the Bellman equation,

$$\langle F_x(\theta_i) \rangle = \langle F_x(\theta_{i-1}) \rangle + \max\langle \Delta F_x(\theta_{i-1}) \rangle. \quad (27)$$

Figures 13(A) and (B) presents the results of two offline parameter planning (i.e. Task 1 starts from s10 and Task 2 starts from s20), respectively. It is worth noting that each stage covers 400 cases, but some cases have already been computed in the previous stage. To avoid duplicate calculations, all computation results are stored.

The results of the offline parameter planning are applied to the 2D prototype of MUPRO introduced

in section 2.1, where each parameter combination is run for 2 motion cycles. The experimental results corresponding to Task 1 are presented in table 3. In Task 1, the relative error between the experimental measurements $\langle F_x \rangle^{\text{exm}}$ and the MPOD-NIROM simulations $\langle F_x \rangle^{\text{MPOD}}$ is around 5%. This indicates that the predicted results of MPOD-NIROM exhibit a consistent trend with the experimental measurements. The computational cost of the proposed MPOD-NIROM for the online prediction process and post-processing (as shown in figure 3) is approximately 48 ms per case. This is about 50% less than the classical POD-NIROM (case3) and approximately four orders of magnitude faster than the HF model.

In Task 2 (detailed in table 4), the wavenumber at the initial stage is reset. However, the undulating period at the initial stage is the same for Task 1 and Task 2, since the undulating period is positively correlated with the propulsion (which was well-studied [5, 7, 13]). Figure 13(C) shows both the offline parameter planning paths of Task 1 and Task 2, as well

Table 4. Results of offline parameter planning (Task 2) based on MPOD-NIROM. The computing environment has been reported in table 3.

Stage	s20	s21	s22	s23	s24
k	0.7	0.6	0.5	0.4	0.33
T	3	2.7	2.4	2.1	1.8
$\langle F_x \rangle^{\text{MPOD}}$ (Nm ⁻¹)	0.701	0.865	1.213	1.730	2.578
Computational cost (ms)	9612	14537	14396	14363	14498
Number of cases calculated	200	300	300	300	300

as the contours of $\langle F_x \rangle^{\text{MPOD}}$. We find that both Task 1 and Task 2 converge at $k = 0.33$. Meanwhile, the contour of $\langle F_x \rangle^{\text{MPOD}}$ shows that a local maximum of propulsion occurs near $k = 0.33$. These results indicate that the proposed framework can effectively plan the undulating parameters of MUPRo to find the maximum propulsive force.

4. Conclusion

In this paper, the MPOD-NIROM framework was developed to predict the instantaneous hydrodynamic forces acting on a undulating mantle. The framework comprises three parts: preprocessing, online prediction, and post-processing. In preprocessing, the features of the solution states are identified at multiple scales based on the proposed MPOD algorithm. The LSTM-based NIROM is then trained to predict the temporal evolution of multi-scale features. During the online prediction process, the trained LSTM-based NIROM predicts the temporal evolution of the fluid's multi-scale features, taking the prescribed mantle motion as input. The instantaneous hydrodynamic forces acting on the mantle are then calculated based on the predicted solution states in the post-processing phase. To efficiently identify fluid features in the solution states within the specified region, we propose the MPOD algorithm. The results show that the proposed MPOD algorithm is promising for local feature extraction. Based on this proposed algorithm, $\text{NMSE}(s_i|_{\Gamma_m})$ converges almost linearly to 0.2% with respect to the DoF or the expansion of the POD mode index. To reliably predict the solution states located around the undulating mantle, i.e. $\text{NMSE}(s_i|_{\Gamma_m}) \leq 0.2\%$, the DoF required by the proposed MPOD algorithm is 46% of that required by the classical POD algorithm.

The hydrodynamic force acting on the undulating mantle, calculated from the predicted solution states, can be predicted economically and reliably. The trained LSTM-based NIROM can predict the temporal evolution of the fluid features extracted from the proposed MPOD algorithm based on the input (i.e. the specified mantle undulation). The results show that the error $\text{NMSE}(F_x)$ of the developed MPOD-NIROM framework in predicting the hydrodynamic forces acting on the mantle with DoF = 204 is comparable to the classical POD-based NIROM with DoF = 444. Moreover, when DoF = 204, the

$\text{NMSE}(F_x)$ of the proposed framework is one or two orders of magnitude lower than that of classical POD-NIROM framework.

The developed MPOD-NIROM framework is promising for the offline parameter planning and online propulsion control of aquatic-inspired robots, since it enables the prediction of hydrodynamic forces acting on the robot economically and reliably. In the offline parameter planning of MUPRo propulsion with 1320 cases, the computational cost of the proposed MPOD-NIROM model is approximately 63.32 s, and the predicted results show consistent trends with the 2D prototype experiments. Future research could focus on the reliability and generality of the developed MPOD-NIROM framework. The reliability could require employing a data assimilation algorithm [65]. At the same time, generality of the developed framework may need to be evaluated based on other cases of aquatic-inspired robots [22, 72]. Meanwhile, the predicted fluid characteristics of the present study may be different from the real phenomena, because of the 2D assumption made judge from the spatial dimension. Thus, predicting the hydrodynamic forces acting on the underwater vehicle from 3D model is also a future scope of this paper.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

Acknowledgments

This work was supported by the National Natural Science Foundation of China (Grant Nos. 51575478 and 61571007).

Appendix A. Details on the HF model

The ALE method is used in presented computations to deal with the fluid-structure interaction problem, and governing equations are as follows,

$$\dot{\mathbf{u}} + (\mathbf{u} \cdot \nabla)(\mathbf{u} - \dot{\mathbf{q}}) = -\frac{1}{\rho_f} \nabla p + \frac{\mu}{\rho_f} \nabla \cdot [\nabla \mathbf{u} + \nabla \mathbf{u}^T] \text{ in } \Omega_f, \quad (\text{A.1})$$

$$\nabla \cdot \mathbf{u} = 0, \text{ in } \Omega_f, \quad (\text{A.2})$$

where $\mathbf{u} = [u, v]$ is the fluid velocity, $\mathbf{q} = [q_x, q_y]$ is the grid displacement, p is the pressure, ρ is the fluid's density, and $\mu = 1.01 \times 10^{-3} \text{Pa} \cdot \text{s}$ is the dynamic viscosity, respectively. The grid displacement $\mathbf{q}$ is input as an independent variable, governed by a modified version of the Laplace grid motion algorithm,

$$\nabla \cdot ([1 + \tau_e] \nabla) \mathbf{q} = 0, \quad (\text{A.3})$$

$$\mathbf{q} = \mathbf{q}_m|_{\Gamma_m}, \quad (\text{A.4})$$

$$\mathbf{q} = \mathbf{0}|_{\Gamma}, \quad (\text{A.5})$$

where Γ_m and Γ represent the moving boundary (corresponding to the mantle) and the fixed boundaries, respectively, τ_e is the discrete weighting parameter defined in previous studies [73, 74], $\mathbf{q}_m$ is the prescribed grid displacement on Γ_m calculated from the displacement on the body axis of the mantle $\mathbf{q}_m(\xi, t) = [0, Y(\xi, t)]$.

When solving equations (A.1) and (A.2) a three-step fractional step method is employed for the time discretization of the system, as described in previous studies [39, 75, 76]. To limit the numerical oscillation of the solution states, the Streamline upwind Petrov–Galerkin finite element method is used for the spatial discretization of the system [77, 78]. The above algorithm has been employed in the numerical simulation of wake structures with upstream moving objects [75, 79]. The current HF model can be summarized in the following steps: (i) update the mantle boundary displacement $\mathbf{q}_m(t^{n+1}) = [0, Y(t^{n+1})]$, by solving the parametric model of the mantle motion (equations (1) and (2)); (ii) update the grid configuration $\mathbf{q}(t^{n+1})$ based on the mesh motion algorithm equations (A.3)–(A.5); (iii) update the solution states $\mathbf{s}(t^{n+1})$ by solving the Navier–Stokes equations (equations (A.1) and (A.2)) on the updated grid configuration $\mathbf{q}(t^{n+1})$; (iv) return to (i).

Appendix B. Details on LSTM NN

The LSTM NN in this paper contains an LSTM layer (with n_s LSTM cells) that describes long-term dependencies between input and output. Expressly, The hidden state h (also known as the output state, which describes the output of the LSTM layer) and the cell state c (which contains the information obtained in the previous time step) undertake the data flow in the LSTM layer [80]. As specified in equations (B.1)–(B.3), the learnable coefficients of the LSTM layer include the input weight W , the recurrent weight R and the bias b

$$W = [W_i, W_f, W_c, W_o]^T, \quad (\text{B.1})$$

$$R = [R_i, R_f, R_c, R_o]^T, \quad (\text{B.2})$$

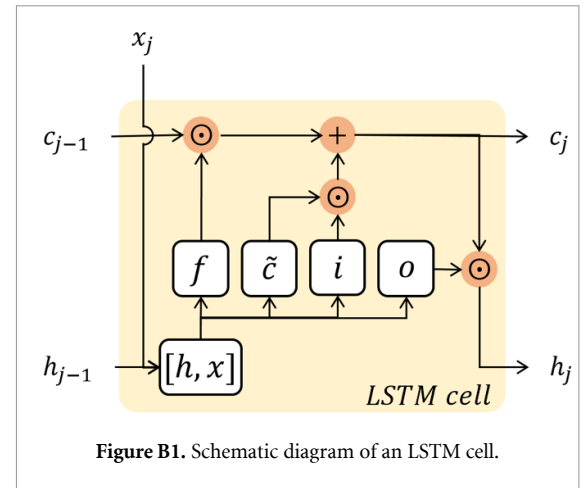


Figure B1. Schematic diagram of an LSTM cell.

$$b = [b_i, b_f, b_c, b_o]^T, \quad (\text{B.3})$$

where $i \in \mathbf{R}^x$ is the input gate, $f \in \mathbf{R}^x$ is the forget gate, $o \in \mathbf{R}^h$ is the output gate, and $\tilde{c} \in \mathbf{R}^c = \mathbf{R}^h$ is the cell candidate gate. Figure B1 illustrates the data flow in the LSTM cell at the time step t_j where the states and the components are introduced in equations (B.4)–(B.9),

$$\text{Cell state } c_j = f_j \odot c_{j-1} + i_j \odot \tilde{c}_j, \quad (\text{B.4})$$

$$\text{Hidden state } h_j = o_j \odot \sigma_c(c_j), \quad (\text{B.5})$$

$$\text{Input gate } i_j = \sigma_g(W_i x_j + R_i h_{j-1} + b_i), \quad (\text{B.6})$$

$$\text{Forget gate } f_j = \sigma_g(W_f x_j + R_f h_{j-1} + b_f), \quad (\text{B.7})$$

$$\text{Cell candidate } \tilde{c}_j = \sigma_c(W_c x_j + R_c h_{j-1} + b_c), \quad (\text{B.8})$$

$$\text{Output gate } o_j = \sigma_g(W_o x_j + R_o h_{j-1} + b_o), \quad (\text{B.9})$$

where $\odot$ denotes the Hadamard product, $\sigma_c(x) = \tanh(x)$ denotes the state activation function, $\sigma_g(x) = (1 + \exp(-x))^{-1}$ denotes the gate activation function.

ORCID iD

Zixiang Ying  <https://orcid.org/0000-0003-0637-9819>

References

- [1] Webb P 1984 Body form, locomotion and foraging in aquatic vertebrates *Int. Compar. Biol.* **24** 107–20
- [2] Nelson J, Grande T and Wilson M 2016 *Fishes of the World* 5th edn (Wiley)
- [3] Videler J, Müller U and Stamhuis E J 2000 Aquatic vertebrate locomotion: wakes from body waves *J. Exp. Biol.* **202** 3423–30

- [4] Lucas K, Lauder G and Tytell E 2020 Airfoil-like mechanics generate thrust on the anterior body of swimming fishes *Proc. Natl Acad. Sci.* **117** 201919055
- [5] Borazjani I and Sotiropoulos F 2008 Numerical investigation of the hydrodynamics of carangiform swimming in the transitional and inertial flow regimes *J. Exp. Biol.* **211** 1541–58
- [6] Gray J 2020 Studies in animal locomotion. I. The movement of fish with special reference to the eel *J. Exp. Biol.* **10** 88–104
- [7] Porez M, Boyer F and Ijspeert A J 2014 Improved Lighthill fish swimming model for bio-inspired robots: modeling, computational aspects and experimental comparisons *Int. J. Robot. Res.* **33** 1322–41
- [8] Tytell E and Lauder G 2004 The hydrodynamics of eel swimming. I. Wake structure *J. Exp. Biol.* **207** 1825–41
- [9] Shadwick R and Syme D 2008 Thunniform swimming: muscle dynamics and mechanical power production of aerobic fibres in yellowfin tuna (*Thunnus albacares*) *J. Exp. Biol.* **211** 1603–11
- [10] Dabiri J, Colin S P, Katija K and Costello J H 2010 A wake-based correlate of swimming performance and foraging behavior in seven co-occurring jellyfish species *J. Exp. Biol.* **213** 1217–25
- [11] Dabiri J, Colin S P, Costello J H and Gharib M 2005 Flow patterns generated by oblate medusan jellyfish: field measurements and laboratory analyses *J. Exp. Biol.* **208** 1257–65
- [12] Hu J, Li H and Chen W 2021 A squid-inspired swimming robot using folding of origami *J. Eng.* **2021** 630–9
- [13] Costello J et al 2020 The hydrodynamics of jellyfish swimming *Annu. Rev. Mar. Sci.* **13** 1–22
- [14] Renda F, Giorgio-Serchi F, Boyer F and Laschi C 2015 Modelling cephalopod-inspired pulsed-jet locomotion for underwater soft robots *Bioinspir. Biomim.* **10** 055005
- [15] Marut K, Stewart C, Michael T, Villanueva A and Priya S 2013 A jellyfish-inspired jet propulsion robot actuated by an iris mechanism *Smart Mater. Struct.* **22** 094021
- [16] Batty R S 1996 Life in moving fluids: the physical biology of flow *J. Exp. Mar. Biol. Ecol.* **198** 153
- [17] Gilewski W, Pelczynski J and Stawarz P 2014 A comparative study of origami inspired folded plates *Proc. Eng.* **91** 220–5
- [18] Peraza Hernandez E, Hartl D and Malak R 2013 Design and numerical analysis of an SMA mesh-based self-folding sheet *Smart Mater. Struct.* **22** 094008
- [19] Roper D et al 2011 A review of developments towards biologically inspired propulsion systems for autonomous underwater vehicles *Proc. Inst. Mech. Eng. M* **225** 77–96
- [20] Baldwin T and Battista N 2021 Hopscotching jellyfish: combining different duty cycle kinematics can lead to enhanced swimming performance *Bioinspir. Biomim.* **16** 066021
- [21] Joshi A, Kulkarni A and Tadesse Y 2019 FludoJelly: experimental study on jellyfish-like soft robot enabled by soft pneumatic composite (SPC) *Robotics* **8** 56
- [22] Frame J, Lopez N, Curet O and Engeberg E D 2018 Thrust force characterization of free-swimming soft robotic jellyfish *Bioinspir. Biomim.* **13** 064001
- [23] Najem J et al 2012 Biomimetic jellyfish-inspired underwater vehicle actuated by ionic polymer metal composite actuators *Smart Mater. Struct.* **21** 094026
- [24] Hoover A, Porras A and Miller L 2019 Pump or coast: the role of resonance and passive energy recapture in medusan swimming performance *J. Fluid Mech.* **863** 1031–61
- [25] Miles J and Battista N 2019 Naut your everyday jellyfish model: exploring how tentacles and oral arms impact locomotion *Fluids* **4** 169
- [26] Hoover A, Griffith B and Miller L 2017 Quantifying performance in the medusan mechanospace with an actively swimming three-dimensional jellyfish model *J. Fluid Mech.* **813** 1112–55
- [27] Lipinski D and Mohseni K 2009 Flow structures and fluid transport for the hydromedusae *Sarsia tubulosa* and *Aequorea victoria* *J. Exp. Biol.* **212** 2436–47
- [28] Dular M et al 2009 Numerical investigation of flow in the vicinity of a swimming jellyfish *Eng. Appl. Comput. Fluid Mech.* **3** 258–70
- [29] Ying Z, Zhang H, Wang L and Melnik R 2022 Propulsion optimization of a jellyfish-inspired robot based on non-intrusive reduced-order model with proper orthogonal decomposition *Bioinspir. Biomim.* **17** 046005
- [30] Jones K and Platzer M 1997 Numerical computation of flapping-wing propulsion and power extraction AIAA **97** 97–0826
- [31] Sirovich L 1987 Turbulence and the dynamics of coherent structures. I—Coherent structures. II—Symmetries and transformations. III—Dynamics and scaling *Q. Appl. Math.* **45** 561–71
- [32] Holmes P, Lumley J and Berkooz G 1996 *Turbulence, Coherent Structures, Dynamical Systems and Symmetry* (Cambridge University Press) (<https://doi.org/10.1017/CBO9780511622700>)
- [33] Schilders W H A, Vorst H A V D and Rommes J 2008 *Model Order Reduction: Theory, Research Aspects and Applications* (Springer)
- [34] Zhao X, Wu C, Chen Z, Yang L and Shu C 2020 Reduced order modeling-based discrete unified gas kinetic scheme for rarefied gas flows *Phys. Fluids* **32** 067108
- [35] Dragomirescu I, Resiga R and Muntean S 2013 Proper orthogonal decomposition method in swirling flows applications *AIP Conf. Proc.* **1558** 1349–52
- [36] Perrin R et al 2006 *3D circular cylinder FLOMANIA — A European Initiative on Flow Physics Modelling (Berlin, Heidelberg)* vol 94 pp 299–312
- [37] Zhao Y, Zhao M, Li X, Liu Z and Du J 2019 A modified proper orthogonal decomposition method for flow dynamic analysis *Comput. Fluids* **182** 28–36
- [38] Erwan L and Hamdouni A 2010 Reduced order modelling method via proper orthogonal decomposition (POD) for flow around an oscillating cylinder *J. Fluids Struct.* **26** 292–311
- [39] Erwan L, Benaouicha M and Hamdouni A 2007 Proper orthogonal decomposition investigation in fluid structure interaction *Eur. J. Comput. Mech./Revue Européenne de Mécanique Numérique* **16** 401–18
- [40] Carere G, Strazzullo M, Ballarin F, Rozza G and Stevenson R 2021 A weighted POD-reduction approach for parametrized PDE-constrained optimal control problems with random inputs and applications to environmental sciences *Comput. Math. Appl.* **102** 261–76
- [41] Kärcher M, Tokoutsis Z, Grepl M A and Veroy K 2018 Certified reduced basis methods for parametrized elliptic optimal control problems with distributed controls *J. Sci. Comput.* **75** 1–32
- [42] Bader E, Grepl M and Veroy K 2017 A certified reduced basis approach for parametrized optimal control problems with two-sided control constraints *Model Reduction of Parametrized Systems* vol 17 (Springer) pp 37–54
- [43] Carlberg K and Farhat C 2011 A low-cost, goal-oriented ‘compact proper orthogonal decomposition’ basis for model reduction of static systems *Int. J. Numer. Methods Eng.* **86** 381–402
- [44] Christensen E, Brøns M and Sørensen J 1999 Evaluation of POD-based decomposition techniques applied to parameter-dependent non-turbulent flows *SIAM J. Sci. Stat. Comput.* **21** 1419–34
- [45] Bistran D and Resiga R 2015 Weighted proper orthogonal decomposition of the swirling flow exiting the hydraulic turbine runner *Appl. Math. Modell.* **40** 4057–78
- [46] Venturi L, Ballarin F and Rozza G 2019 A weighted POD method for elliptic PDEs with random inputs *J. Sci. Comput.* **81** 136–53

- [47] Allery C, Guerin S, Hamdouni A and Sakout A 2004 Experimental and numerical POD study of the Coanda effect used to reduce self-sustained tones *Mech. Res. Commun.* **31** 105–20
- [48] Allery C, Beghein C and Hamdouni A 2005 Applying proper orthogonal decomposition to the computation of particle dispersion in a ventilated room *Commun. Nonlinear Sci. Numer. Simul.* **10** 907–20
- [49] Gunzburger M, Peterson J and Shadid J 2007 Reduced-order modeling of time-dependent PDEs with multiple parameters in the boundary data *Comput. Methods Appl. Mech. Eng.* **196** 1030–47
- [50] Leblond C and Allery C 2014 A priori space–time separated representation for the reduced order modeling of low Reynolds number flows *Comput. Methods Appl. Mech. Eng.* **274** 264–88
- [51] Mohan A T and Gaitonde D V 2018 A deep learning based approach to reduced order modeling for turbulent flow control using LSTM neural networks (arXiv:1804.09269)
- [52] Ying Z, Wang L and Melnik R 2022 Parameter optimization of the bio-inspired robot propulsion through the deep learning based reduced order fluid-structure interaction model *Ocean Eng.* **255** 111436
- [53] Baldi P and Hornik K 1989 Neural networks and principal component analysis: learning from examples without local minima *Neural Netw.* **2** 53–58
- [54] Bourlard H and Kamp Y 1988 Auto-association by multilayer perceptrons and singular value decomposition *Biol. Cybern.* **59** 291–4
- [55] Srinivasan P A, Guastoni L, Azizpour H, Schlatter P and Vinuesa R 2019 Predictions of turbulent shear flows using deep neural networks *Phys. Rev. Fluids* **4** 054603
- [56] Yang M and Xiao Z 2020 Improving the $k-\omega-\gamma$ -Ar transition model by the field inversion and machine learning framework *Phys. Fluids* **32** 064101
- [57] Tu J et al 2013 On dynamic mode decomposition: theory and applications *J. Comput. Dyn.* **1** 391–421
- [58] Alawadhi E 2013 Numerical simulation of fluid flow past an oscillating triangular cylinder in a channel *J. Fluids Eng.* **135** 041202
- [59] De A and Dalal A 2006 Numerical simulation of unconfined flow past a triangular cylinder *Int. J. Numer. Methods Fluids* **52** 801–21
- [60] Lighthill M 1960 Note on the swimming of slender fish *J. Fluid Mech.* **9** 305–17
- [61] Hecht F 2012 New development in FreeFem++ *J. Numer. Math.* **20** 251–66
- [62] Lumley J 1967 The structure of inhomogeneous turbulence *Atmos. Turbul. Radio Wave Propag.* 166–78
- [63] Shinde V, Longatte E, Baj F, Hoarau Y and Braza M 2019 Galerkin-free model reduction for fluid-structure interaction using proper orthogonal decomposition *J. Comput. Phys.* **396** 579–95
- [64] Long J H, Koob T, Schaefer J, Summers A, Bantilan K, Grotmol S and Porter M 2011 Inspired by sharks: a biomimetic skeleton for the flapping, propulsive tail of an aquatic robot *Mar. Technol. Soc. J.* **45** 119–29
- [65] Graybill P, Gluckman B and Kiani M 2022 Optimization of an unscented Kalman filter for an embedded platform *Comput. Biol. Med.* **146** 105557
- [66] Griebel M and Harbrecht H 2019 Singular value decomposition versus sparse grids: refined complexity estimates *IMA J. Numer. Anal.* **39** 1652–71
- [67] Griebel M and Harbrecht H 2014 Approximation of bi-variate functions: singular value decomposition versus sparse grids *IMA J. Numer. Anal.* **34** 28–54
- [68] Borazjani I and Sotiropoulos F 2009 Numerical investigation of the hydrodynamics of anguilliform swimming in the transitional and inertial flow regimes *J. Exp. Biol.* **212** 576–92
- [69] Videler J and Hess F 1984 Fast continuous swimming of two pelagic predators, saithe (*Pollachius virens*) and mackerel (*Scomber scombrus*): a kinematic analysis *J. Exp. Biol.* **109** 209
- [70] Miles J G and Battista N A 2021 Exploring the sensitivity in jellyfish locomotion under variations in scale, frequency, and duty cycle *J. Math. Biol.* **83** 56
- [71] Bellman R E 1962 Dynamic programming treatment of the travelling salesman problem *J. ACM* **9** 61–63
- [72] Marchese A, Onal C D and Rus D 2014 Autonomous soft robotic fish capable of escape maneuvers using fluidic elastomer actuators *Soft Robot.* **1** 75–87
- [73] Masud A, Bhanabagwanwala M and Khurram R A 2007 An adaptive mesh rezoning scheme for moving boundary flows and fluid–structure interaction *Comput. Fluids* **36** 77–91
- [74] Masud A and Hughes T 1997 A space-time Galerkin/least-squares finite element formulation of the Navier-Stokes equations for moving domain problems *Comput. Methods Appl. Mech. Eng.* **146** 91–126
- [75] Bao Y, Zhou D, Huang C, Wu Q and Chen X-Q 2011 Numerical prediction of aerodynamic characteristics of prismatic cylinder by finite element method with Spalart–Allmaras turbulence model *Comput. Struct.* **89** 325–38
- [76] Zhao M, Cheng L, Teng B and Dong G 2007 Hydrodynamic forces on dual cylinders of different diameters in steady currents *J. Fluids Struct.* **23** 59–83
- [77] Choi H G and Yoo J 1997 A fractional four-step finite element formulation of the unsteady incompressible Navier-Stokes equations using SUPG and linear equal-order element methods *Comput. Methods Appl. Mech. Eng.* **143** 333–48
- [78] Brooks A N and Hughes T J R 1982 Streamline upwind/Petrov-Galerkin formulations for convection dominated flows with particular emphasis on the incompressible Navier-Stokes equations *Comput. Methods Appl. Mech. Eng.* **32** 199–259
- [79] Wang H, Yang W, Nguyen K D and Yu G 2014 Wake-induced vibrations of an elastically mounted cylinder located downstream of a stationary larger cylinder at low Reynolds numbers *J. Fluids Struct.* **50** 479–96
- [80] Hochreiter S and Schmidhuber J 1997 Long short-term memory *Neural Comput.* **9** 1735–80